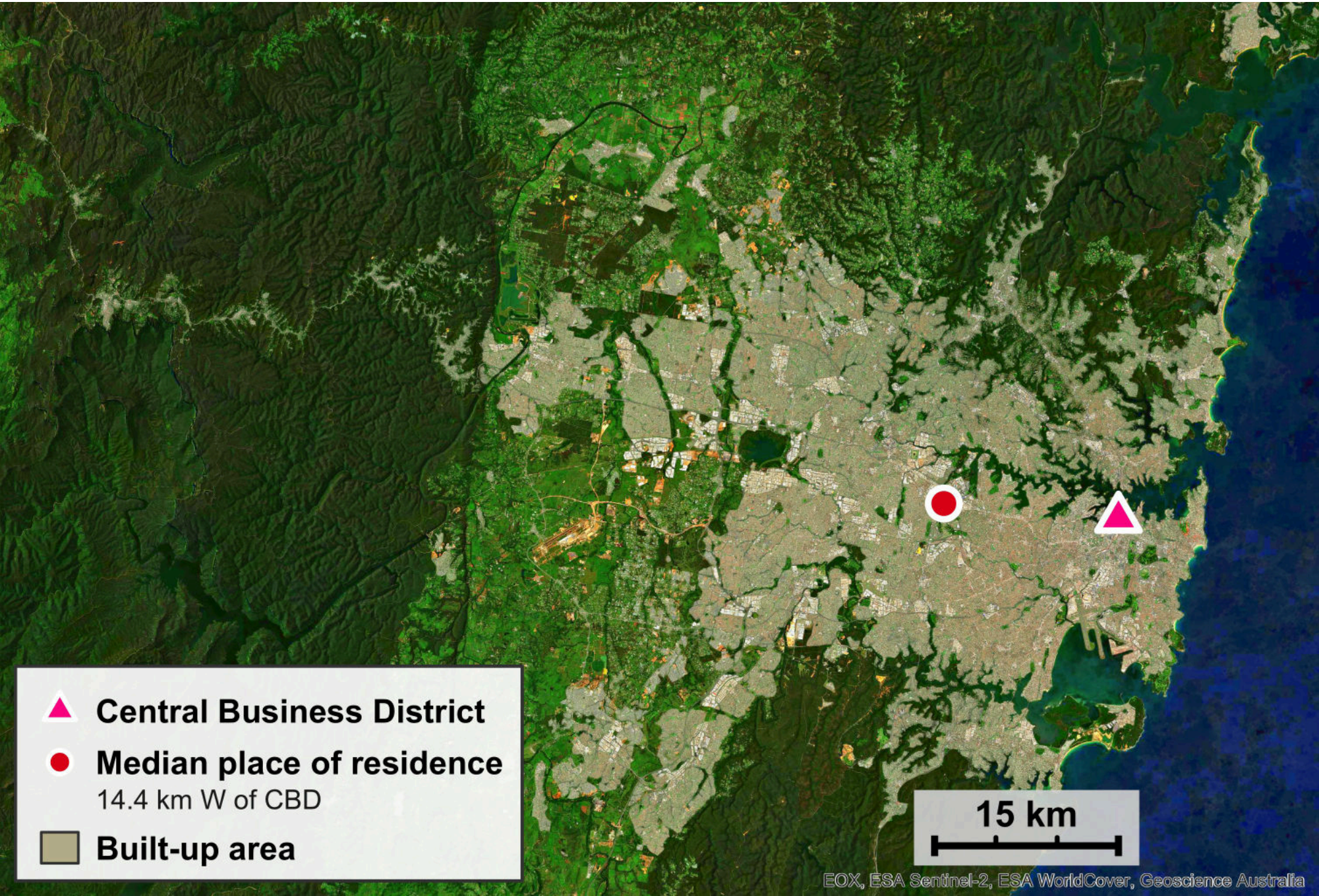
Lopsided Cities

The costs of lopsided jobs distributions, and place-based policy responses

Xavier Tierney

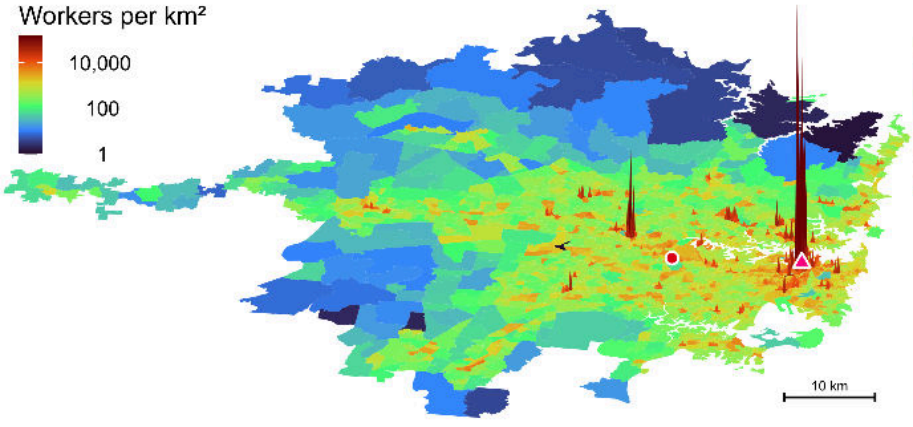
London School of Economics

Sydney: Satellite Image

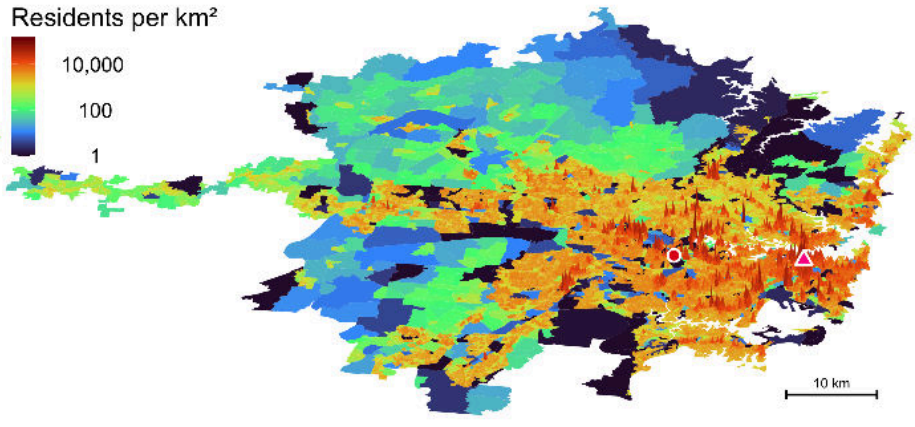


Employment is lopsided

Jobs per km^2



Residents per km^2



One shared colour scale and one shared spike-height scale. 2021 Census: jobs by destination zone, residents by SA1.

Central Business District



Center by Residence



This paper

- **Question:** What are the costs of a lopsided jobs distribution, and can a place based policy counteract them?

Income & education maps →

This paper

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Income & education maps →

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  $\Rightarrow$ output +11.0%; welfare +2.5%; commute -2.1%

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  - Model with public sector and workers split by education
  - Model counterfactual: in so far as the policy is locally successful, it reduces citywide output but may reduce inequality
  - Reduced Form: local commercial land values +50% vs a synthetic control; no private-job response yet

Income & education maps $\rightarrow$

Related literature

- **City shape and geography:** Harari 2020; Saiz 2008; Lee & Lin 2018; Koster 2024; Hsiao 2024

  → *the cost of a lopsided jobs distribution within a city, and its channels*

- **Quantitative urban models:** Ahlfeldt, Redding, Sturm & Wolf 2015; Heblich, Redding & Sturm 2020; Tsivanidis 2026; Kleinman, Liu & Redding 2023; Takeda & Yamagishi 2026

  → *study jobs distribution and public jobs move*

- **Public-sector relocation:** Faggio & Overman 2014; Faggio 2019; Becker, Heblich & Sturm 2021; Nathan, Overman, Riom & Sanchez-Vidal 2024; Faggio, Schlüter & vom Berge 2025; Freitas 2025

  → *a move within one metro; model based welfare incidence estimates*

Roadmap

1. Model explanation and calibration

2. Benchmarking the costs: CBD shift

3. Model of a public-sector move

4. Reduced-form evidence

Model: Workers choose where to live and where to work

Worker o living in i, working in j:

$$u_{ijo} = \frac{B_i \, w_j \, z_{ijo}}{Q_i^{1-\beta} \, e^{\kappa \tau_{ij}}}$$

- B_i residential amenity, Q_i rent price, $1 - \beta$ housing share
- τ_{ij} travel time (minutes), κ commute cost parameter
- z_{ijo} taste draw, Fréchet with dispersion ε

Model: Workers choose where to live and where to work

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$$\text{Share choosing } i,j \;=\; \frac{\left(\dfrac{B_i \, w_j}{Q_i^{1-\beta} \, e^{\kappa \tau_{ij}}} \right)^{\varepsilon}}{\sum_{r,s} \left(\dfrac{B_r \, w_s}{Q_r^{1-\beta} \, e^{\kappa \tau_{rs}}} \right)^{\varepsilon}}, \qquad \nu = \varepsilon \kappa$$

Model: Firms, builders and spillovers

- **Firms:** Produce with labour L_{Fj} and floorspace H_{Fj}, with productivity A_j

$$Y_j = A_j L_{Fj}^{\alpha} H_{Fj}^{1-\alpha}$$

Model: Firms, builders and spillovers

- **Firms:** Produce with labour L_{Fj} and floorspace H_{Fj}, with productivity A_j

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- **Builders:** $H_i = \psi_i Q_i^{\zeta}$, construction productivity ψ_i, supply elasticity ζ

Model: Firms, builders and spillovers

- **Firms:** Produce with labour L_{Fj} and floorspace H_{Fj}, with productivity A_j

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- **Builders:** $H_i = \psi_i \, Q_i^{\zeta}$, construction productivity ψ_i, supply elasticity ζ

- **Spillovers:** Nearby jobs raise productivity, nearby residents raise amenity

$$A_j = a_j \left(\sum_s e^{-\delta \tau_{js}} \frac{L_{Fs}}{K_s} \right)^{\lambda} \qquad B_i = b_i \left(\sum_s e^{-\rho \tau_{is}} \frac{L_{Rs}}{K_s} \right)^{\eta}$$

- L_F jobs, L_R residents
- λ, η spillover strength; δ, ρ decay with travel time
- a_j, b_i: each place's fundamentals

Data and parameters

- **Residents, jobs and incomes:** 2016 Census, aggregated to 4,965 hexagons (400 m)
- **Land values:** valuations for land tax, parcel level, 2016
- **Travel times:** fastest travel time over the 2016 road network
- **Housing share** $1 - \beta = 0.30$: household expenditure survey
- **Spillovers** $\lambda,\ \delta,\ \eta,\ \rho = .07,\ .36,\ .15,\ .76$: Ahlfeldt et al. (2015)
- **Floorspace supply elasticity** $\zeta = 1.86$: Combes, Duranton & Gobillon (2021); consistent with Sydney's land value–rent elasticity

Calibration: Invert the model at Sydney's 2016 equilibrium

- **Commuting decay** ν = 0.126: regressing 2016 commuting flows on travel time
- **Taste dispersion** ε = 6.2: dispersion of observed vs. adjusted wages
- **Invert the equilibrium conditions to calibrate fundamentals:**
  - many residents despite high rents and poor job access $\Rightarrow$ high amenity
  - many workers despite high rents, wages and poor worker access $\Rightarrow$ high productivity

Equations →

Gravity fit →

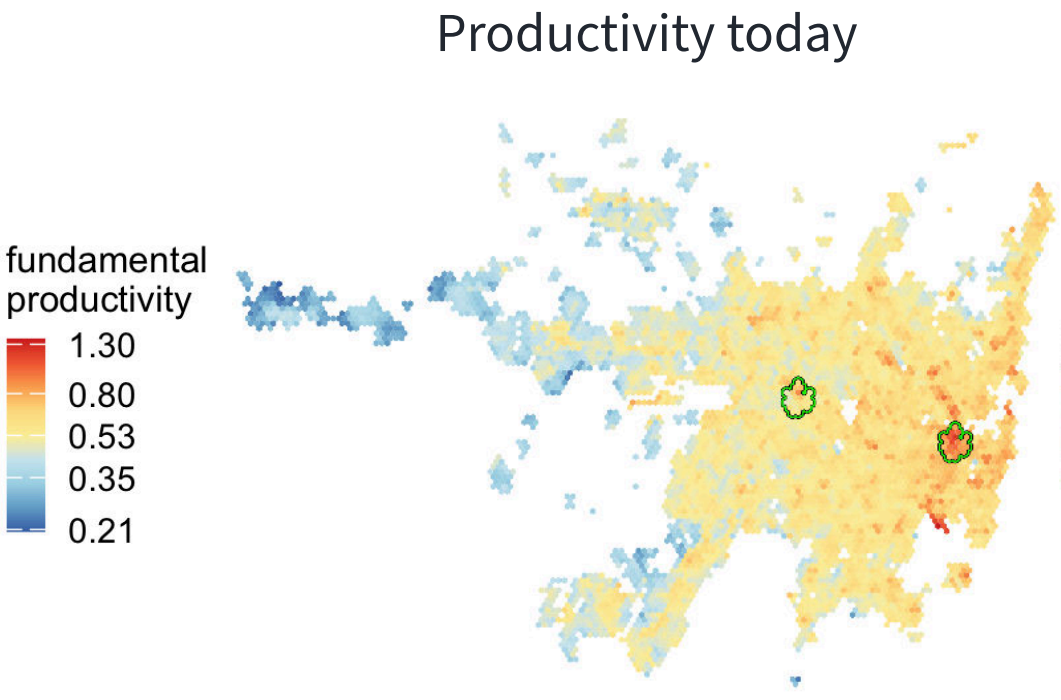
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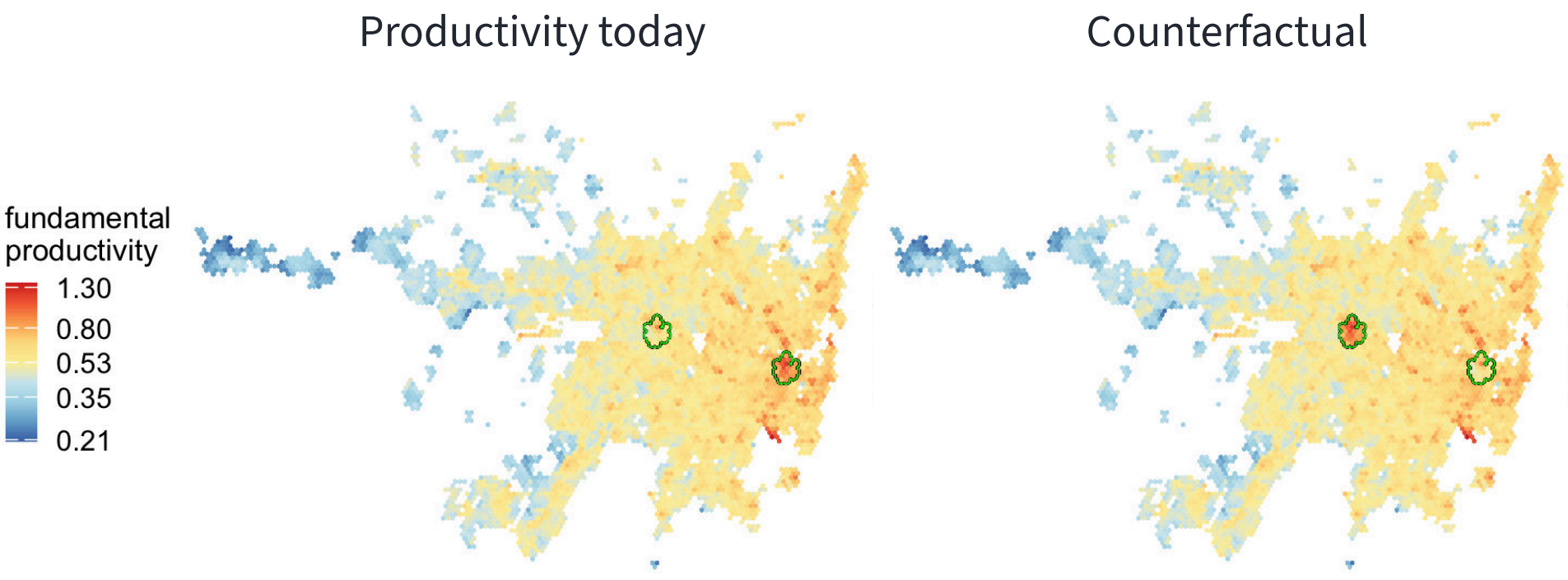
4. Reduced-form evidence

Counterfactual: Move the CBD's fundamentals 20 km west



Fundamental productivity a_j, 400 m hexagons, one shared scale: pale = typical location, red above, blue below; green outlines: the swapped sets.

Counterfactual: Move the CBD's fundamentals 20 km west



→ swap the CBD's 9 km² with Parramatta's: productivity, amenity, land, construction

Fundamental productivity a_j, 400 m hexagons, one shared scale: pale = typical location, red above, blue below; green outlines: the swapped sets. Roads do not move.

Impacts increase as we allow more GE responses

Responses allowed	Δ welfare	Δ output	Δ commute
Jobs move; nobody changes home or job	−3.67%	0.00%	+12.02%

Each row adds one response to the row above; the last row is the headline.

Two-type model →

Robustness →

Impacts increase as we allow more GE responses

Responses allowed	Δ welfare	Δ output	Δ commute
Jobs move; nobody changes home or job	−3.67%	0.00%	+12.02%
+ people re-choose home and job	−0.17%	+3.50%	−0.90%

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Two-type model →

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+ building supply responds	+1.16%	+4.88%	−0.68%

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+ productivity spillovers	+1.25%	+7.06%	−0.05%

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+ productivity spillovers	+1.25%	+7.06%	−0.05%
+ amenity spillovers, half strength	+1.60%	+8.19%	−0.35%

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Two-type model →

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+ amenity spillovers, half strength	+1.60%	+8.19%	−0.35%
+ amenity spillovers, full strength	+2.49%	+10.99%	−2.06%

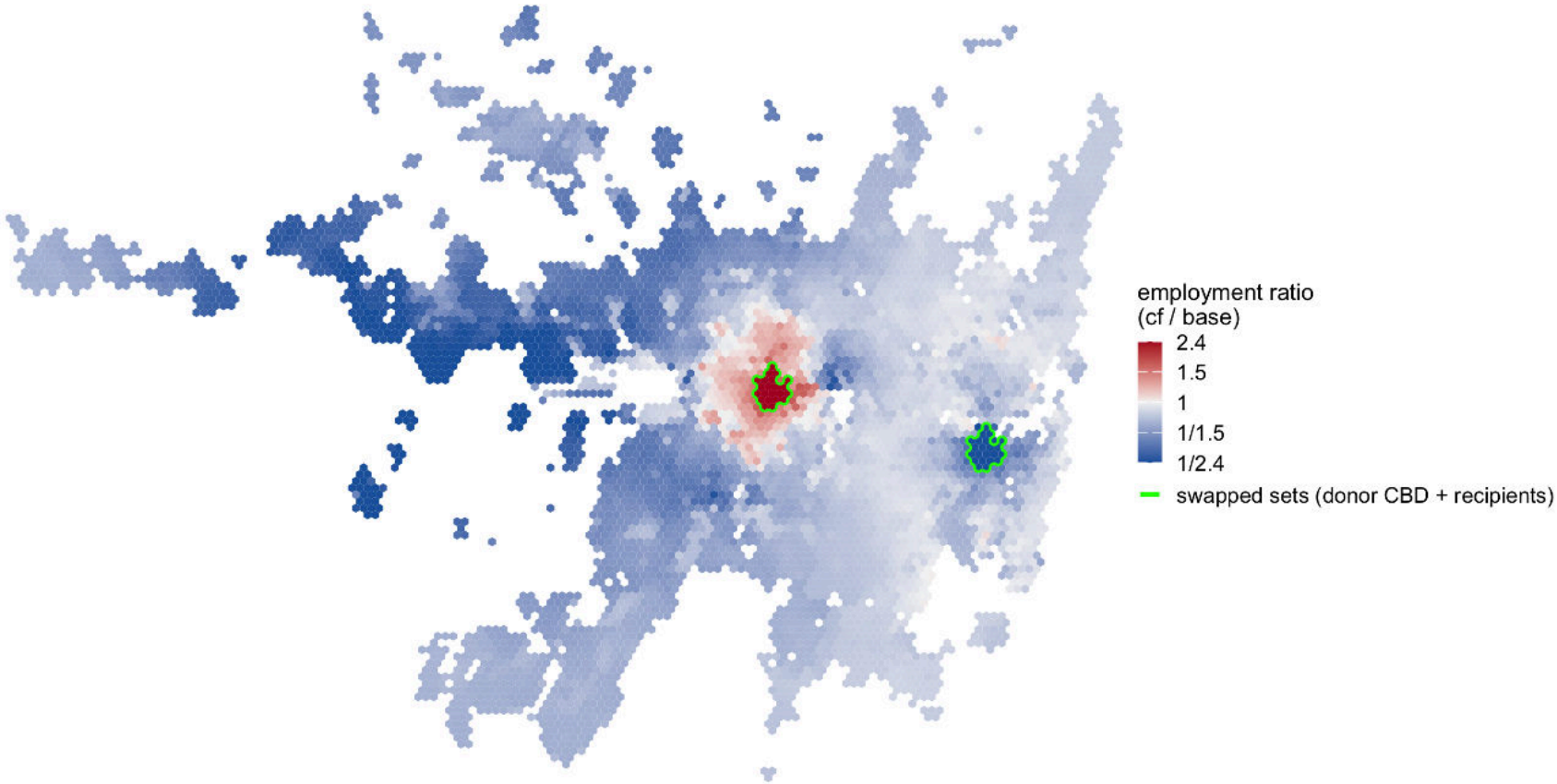
→ when the most productive location is more accessible, output and welfare increase

Each row adds one response to the row above; the last row is the headline.

Two-type model →

Robustness →

Jobs re-form around Parramatta and drain from the rest of the city



Counterfactual / baseline employment per hexagon, log scale. Outside the swapped sets jobs fall 20%; 4.5% of hexagons gain.

Concentration →

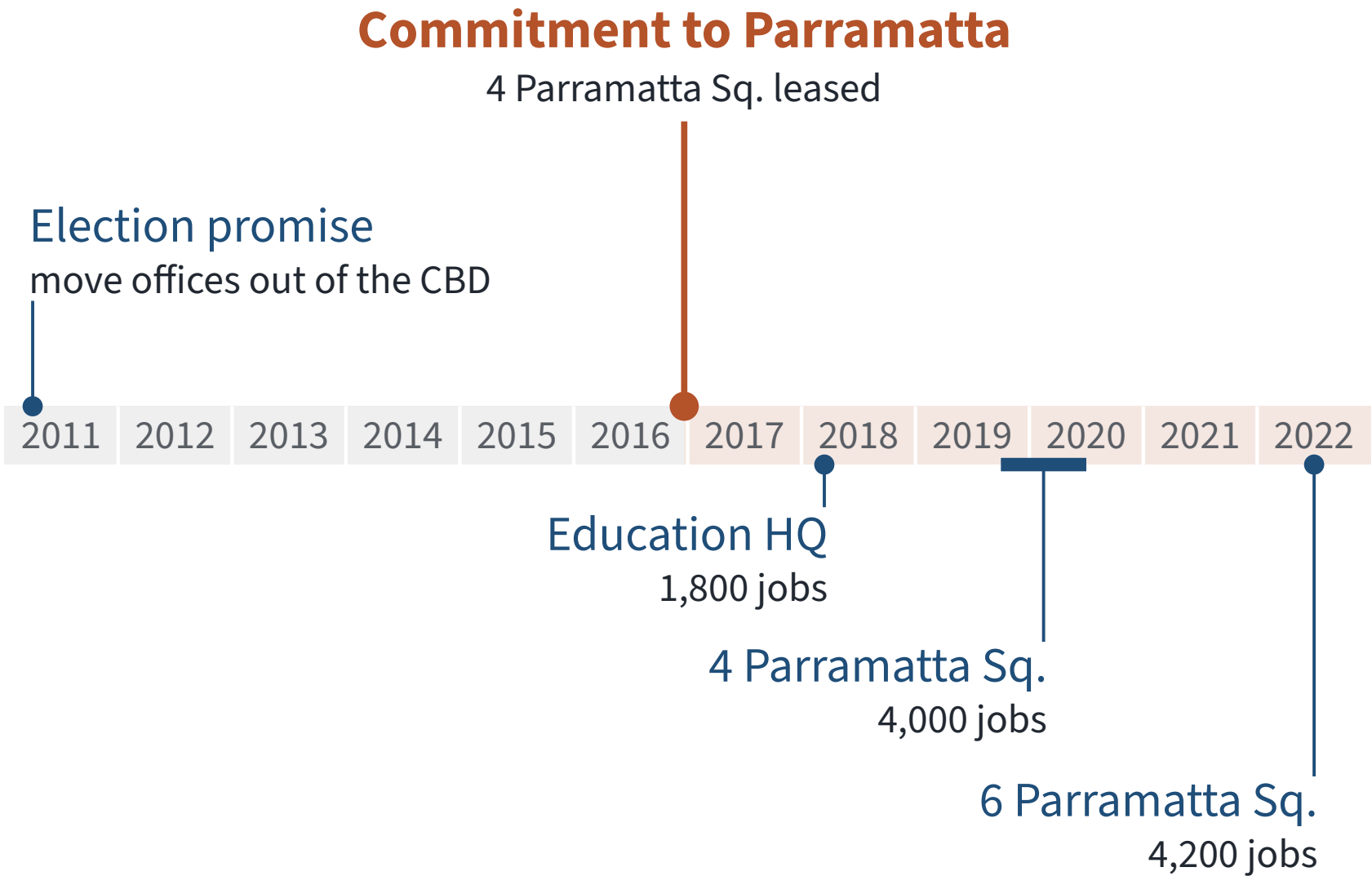
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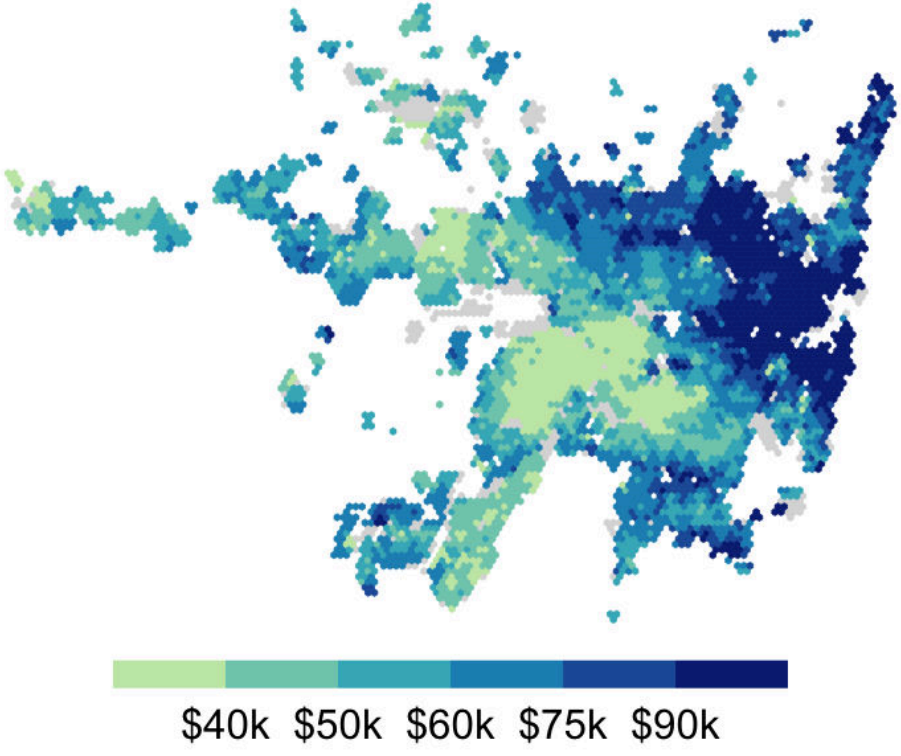
4. Reduced-form evidence

The policy: ∼10,000 public-sector jobs moved to Parramatta, 2018–22

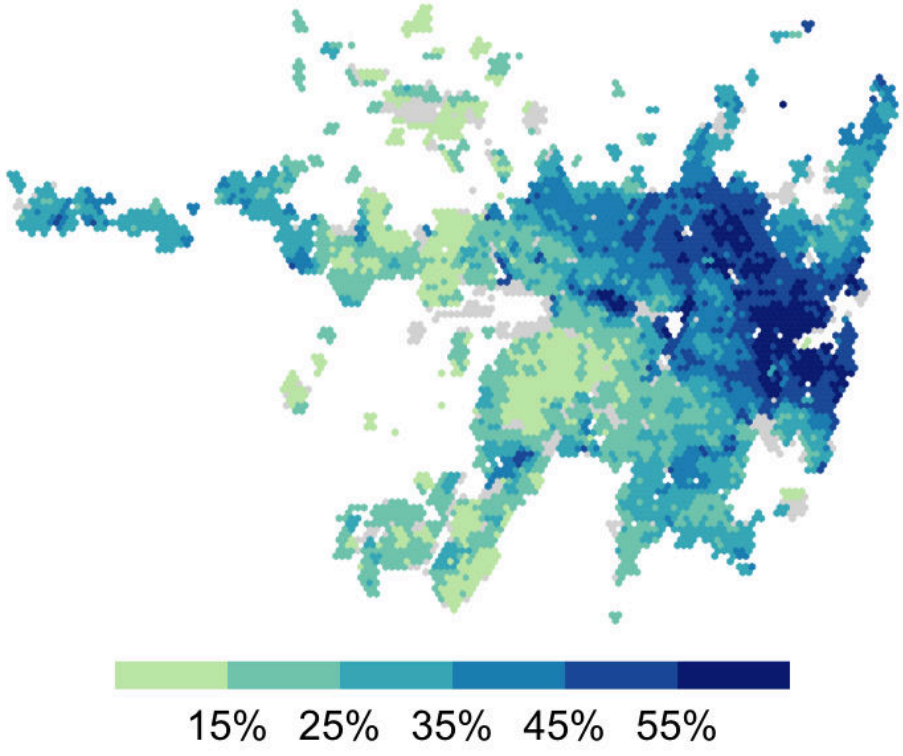


High earners and graduates live in the east

Mean personal income



Share of graduates



By home location, residents aged 15+, on the model's 400 m hexagons. Income: residents with a positive income. Graduates: bachelor's degree or higher. Grey: under 20 residents. 2016 Census.

Adding a public sector and worker types to the model

- **Public sector:** government places its offices
  - wages set centrally; offices rent floorspace like any firm
- **Two types** t: graduates (h), non-graduates (l); own amenities, commuting costs, housing shares
- **One firm per location** hires both types:

$$Y_j = L_{Fj}^{\alpha} H_{Fj}^{1-\alpha}, \qquad L_{Fj} = \left[\sum_t \left((A_j^t)^{1/\alpha} L_{Fj}^t \right)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

A_j^t: productivity of type-t workers L_{Fj}^t; $\sigma = 1.4$: substitution elasticity

$\rightarrow$ *more productive graduates lift nearby non-graduates' wages*

The key unknown: do public workers make nearby private firms more productive?

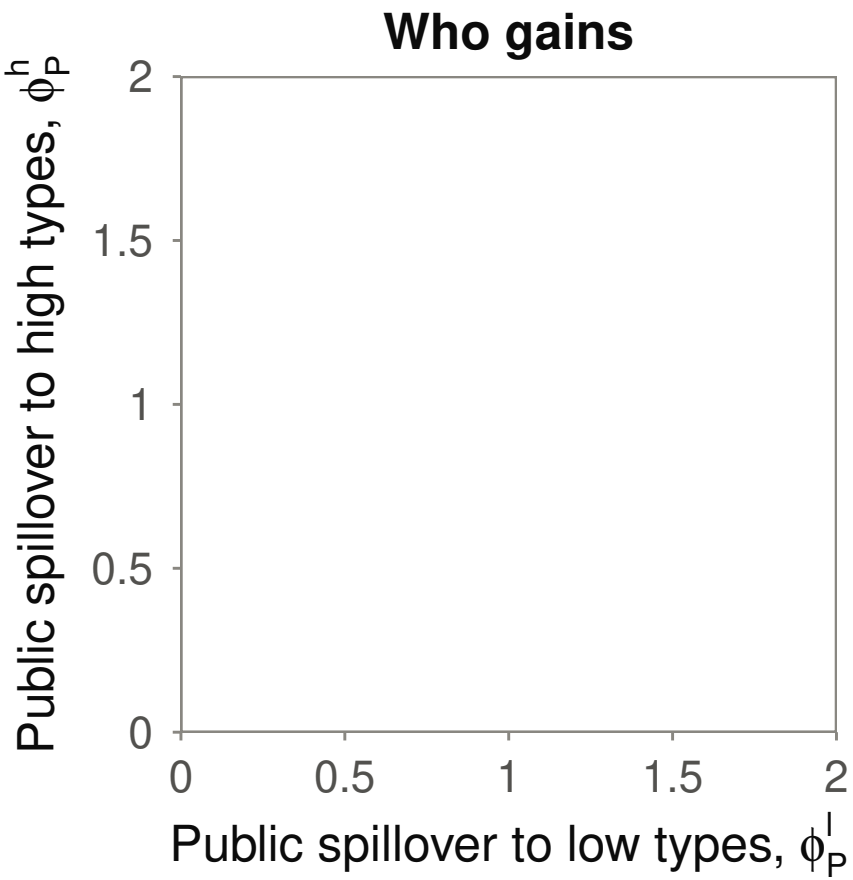
$$A_j^t = a_j^t \left(\sum_s e^{-\delta \tau_{js}} \frac{L_{Fs} + \phi_P^t L_{PFs}}{K_s} \right)^{\lambda}$$

- t: graduate (h) or non-graduate (l) firms; L_F: private jobs; L_{PF}: public jobs
- ϕ_P^t: weight a type-t firm puts on a public neighbour, relative to a private one

$\phi_P^t = 0$: public offices are just tenants
$\phi_P^t = 1$: a public worker is as good a neighbour as a private one

$\rightarrow$ *no direct estimate exists: I show results across* $\phi_P^h, \phi_P^l \in [0, 2]$

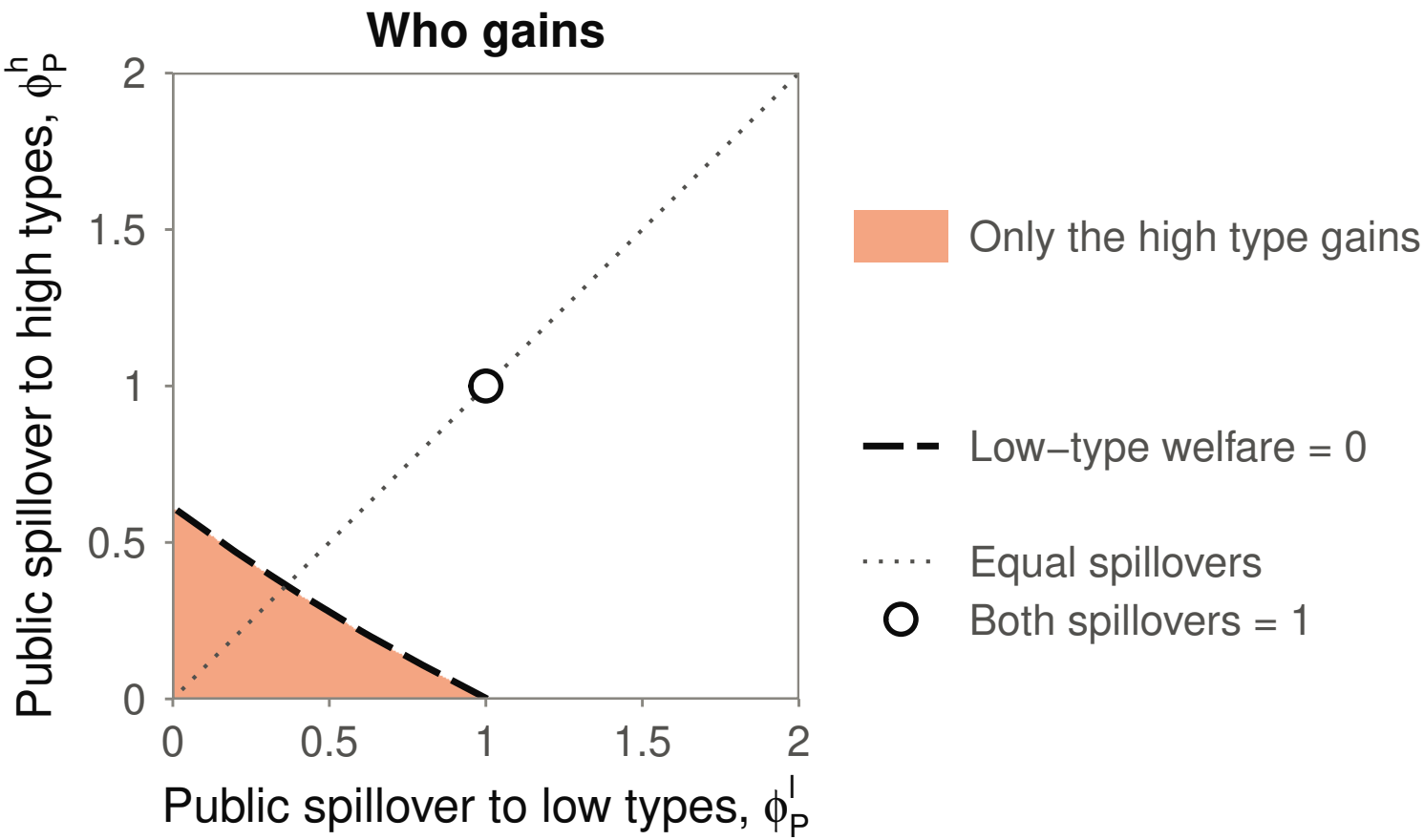
Who gains depends on the public→private spillover



Which half →

Levels →

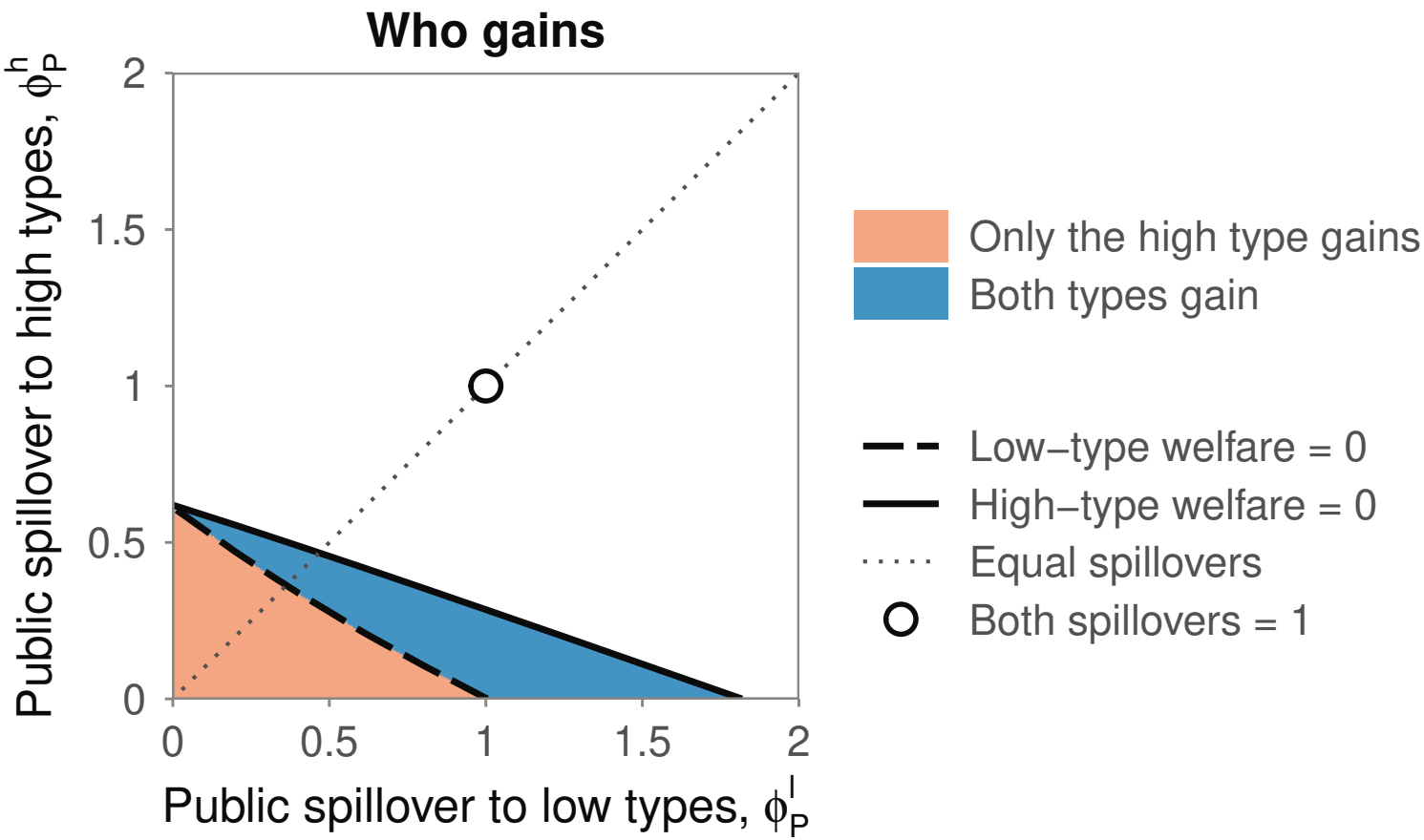
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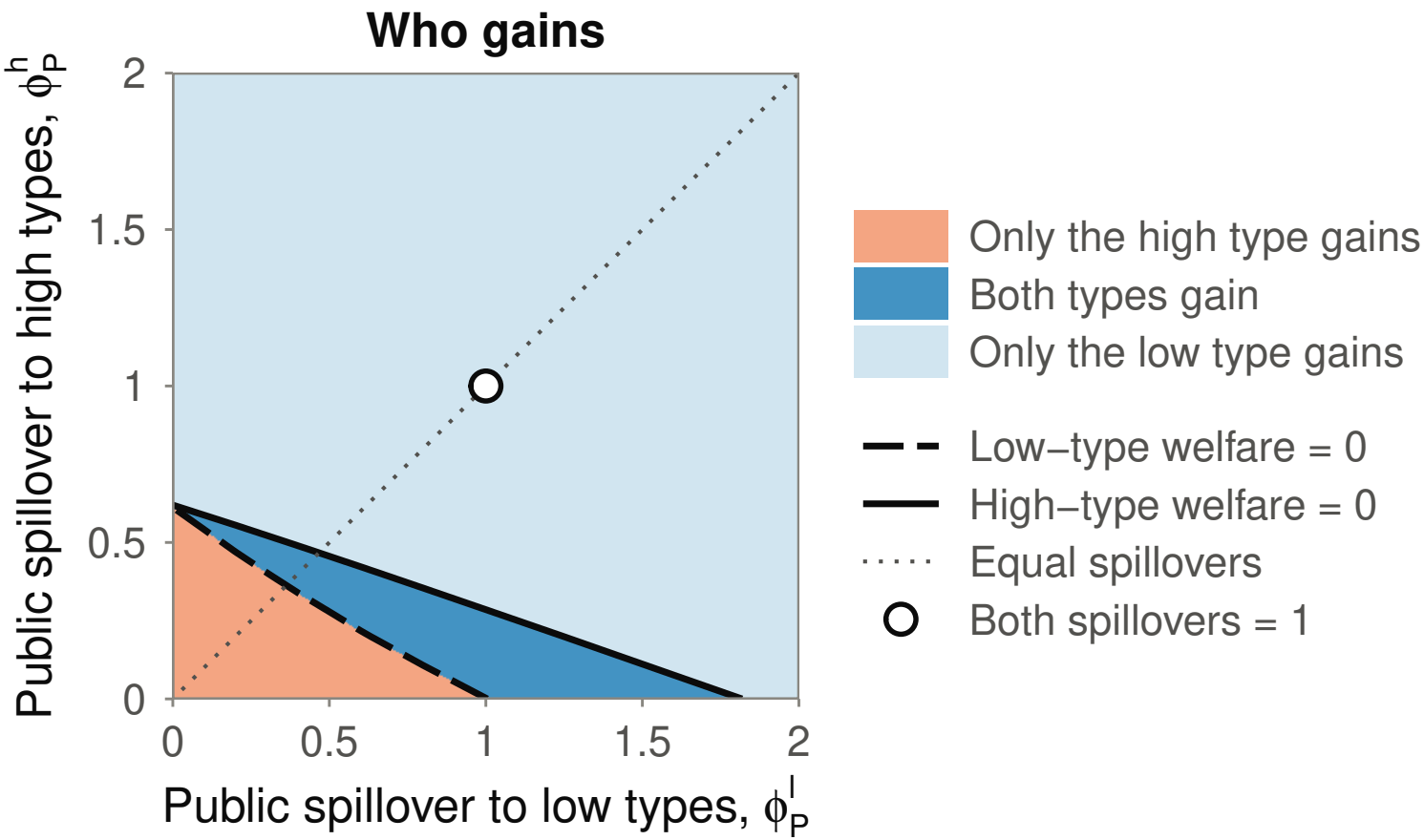
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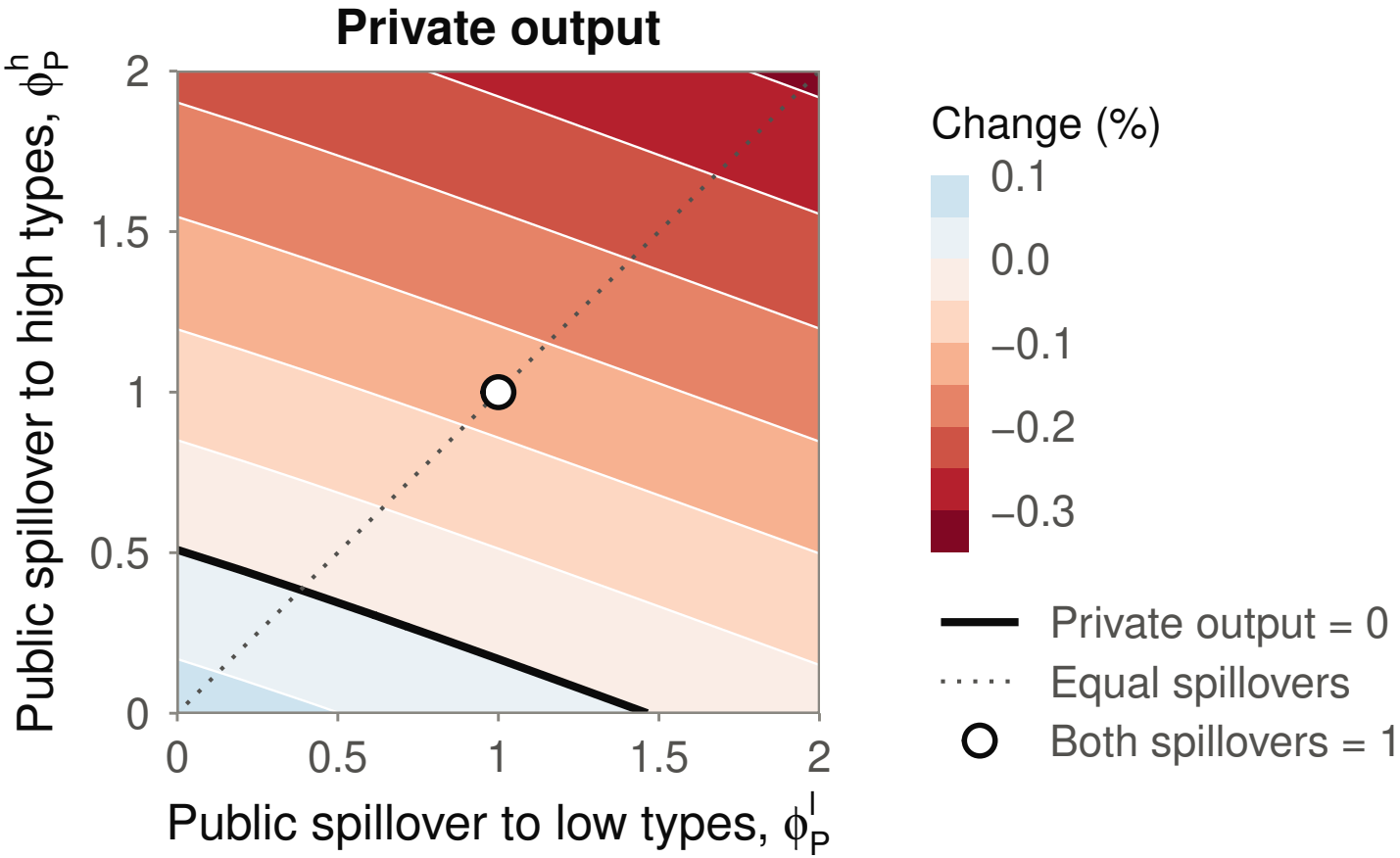
Who gains depends on the public→private spillover



Which half →

Levels →

Who gains depends on the public→private spillover



→ *at* $\phi_P^h = \phi_P^l = 1$*: graduates* −0.06%, *non-graduates* +0.04%, *output* −0.12%

Which half →

Levels →

1. Model explanation and calibration

2. Benchmarking the costs: CBD shift

3. Model of a public-sector move

4. Reduced-form evidence

Synthetic control: data and outcome

- **Data:** the universe of NSW land values (Valuer General), 2012–25
- **Outcome:** land value per m^2 in 800 m-radius circles around each centre
- Calculated from a fixed panel of commercially zoned parcels

Synthetic control: the estimator

- **Synthetic Parramatta:** a weighted blend of the other centres j

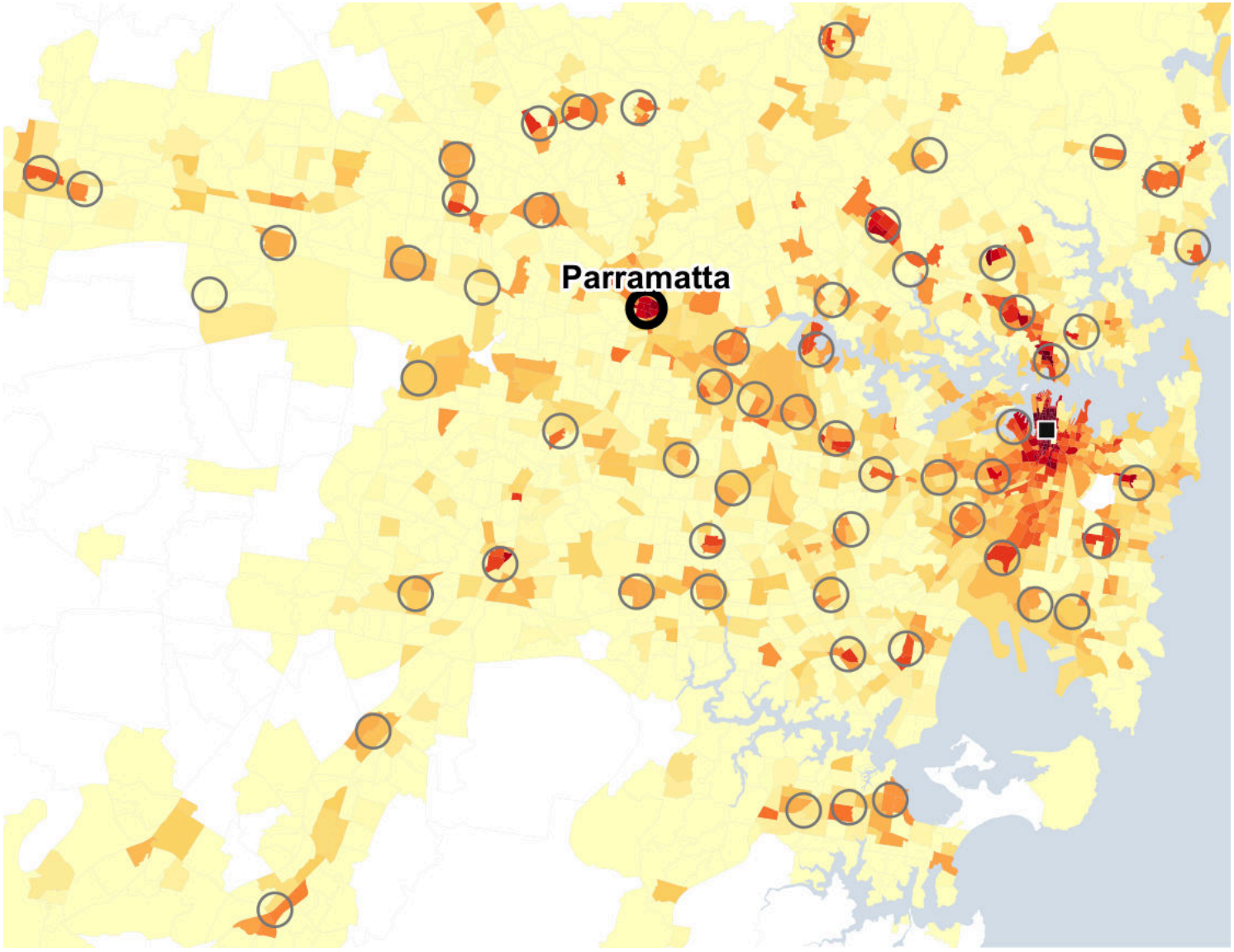
$$Y_t^{\text{synth}}(w) = \sum_j w_j \, Y_{jt}, \qquad w_j \geq 0, \quad \sum_j w_j = 1$$

- **Fit:** weights minimise the gap over the pre-period, 2012–16

$$w^* = \arg\min_w \sum_{t=2012}^{2016} \left(Y_{Pt} - Y_t^{\text{synth}}(w) \right)^2$$

- **Tie-break $\hat{w}$:** the most similar donors, within 1 pp of level of the best fit
- **Effect:** the gap after the Dec 2016 lease, $\hat{\alpha}_t = Y_{Pt} - Y_t^{\text{synth}}(\hat{w})$, $t \geq 2017$
- **Placebo inference:** each donor as if treated — is Parramatta's gap extreme?

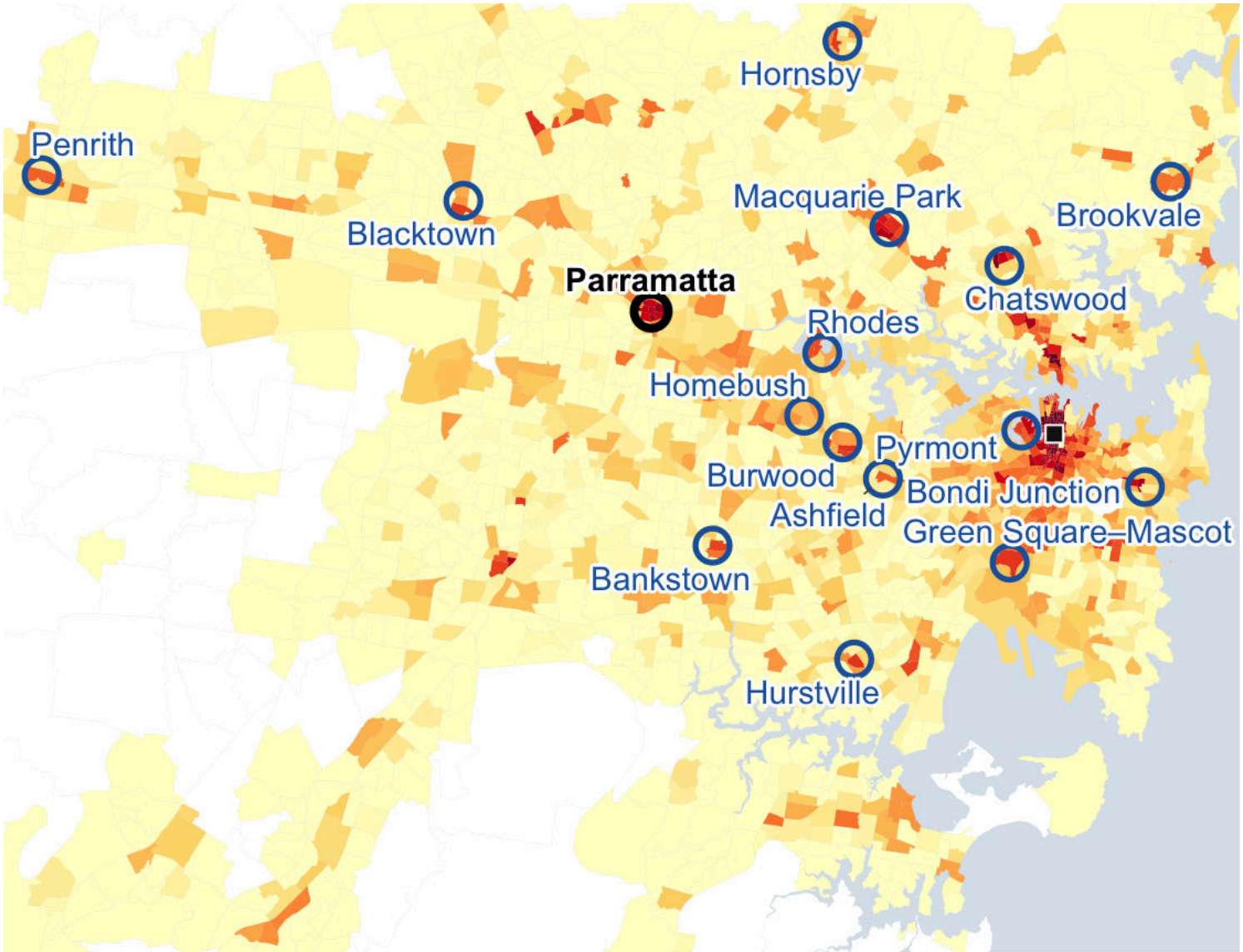
The candidates: Sydney's employment centres



800 m circles over 2016 jobs per km^2; ■ = CBD.

The screens →

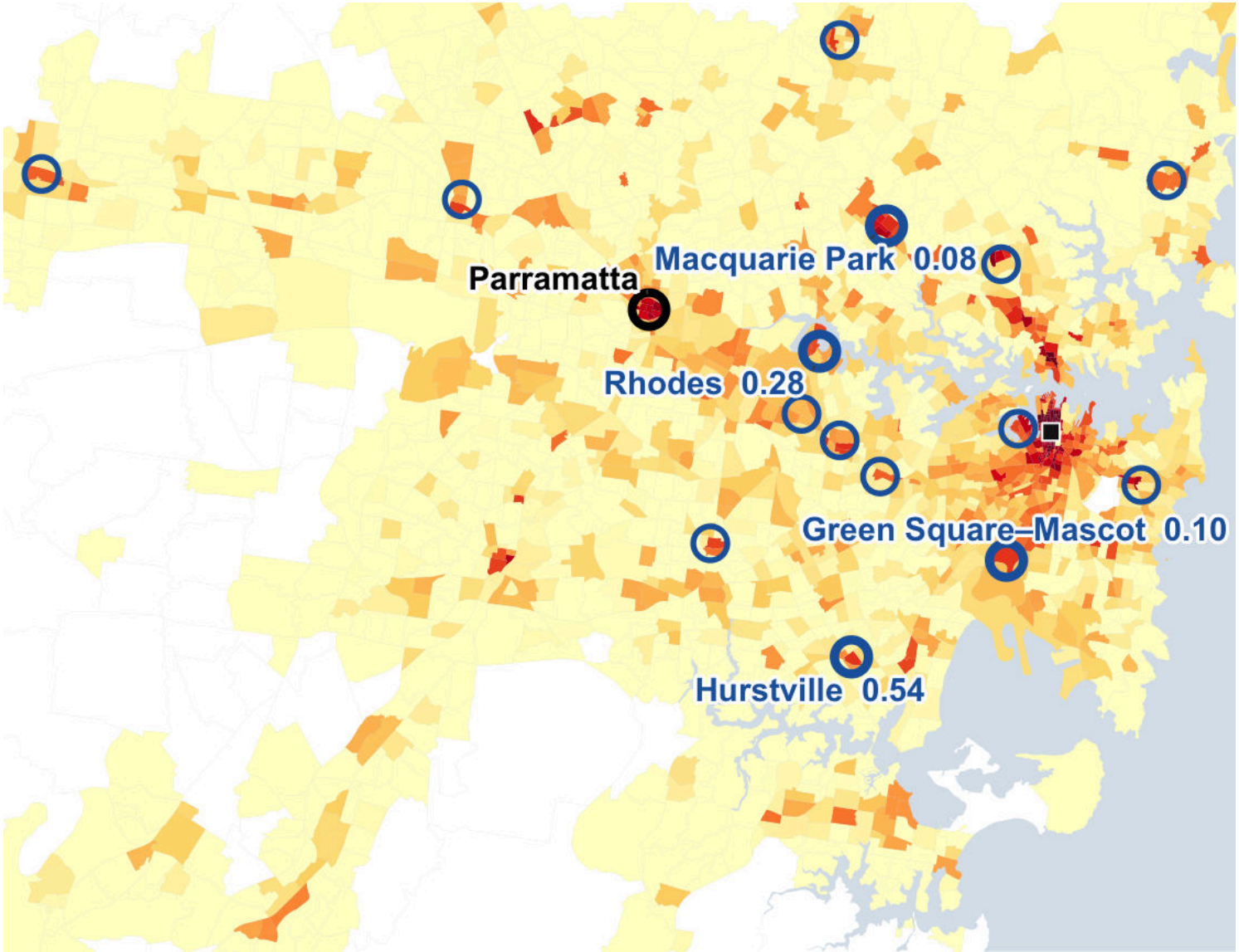
Choose the 15 most similar centres



800 m circles over 2016 jobs per km^2; ■ = CBD.

The screens →

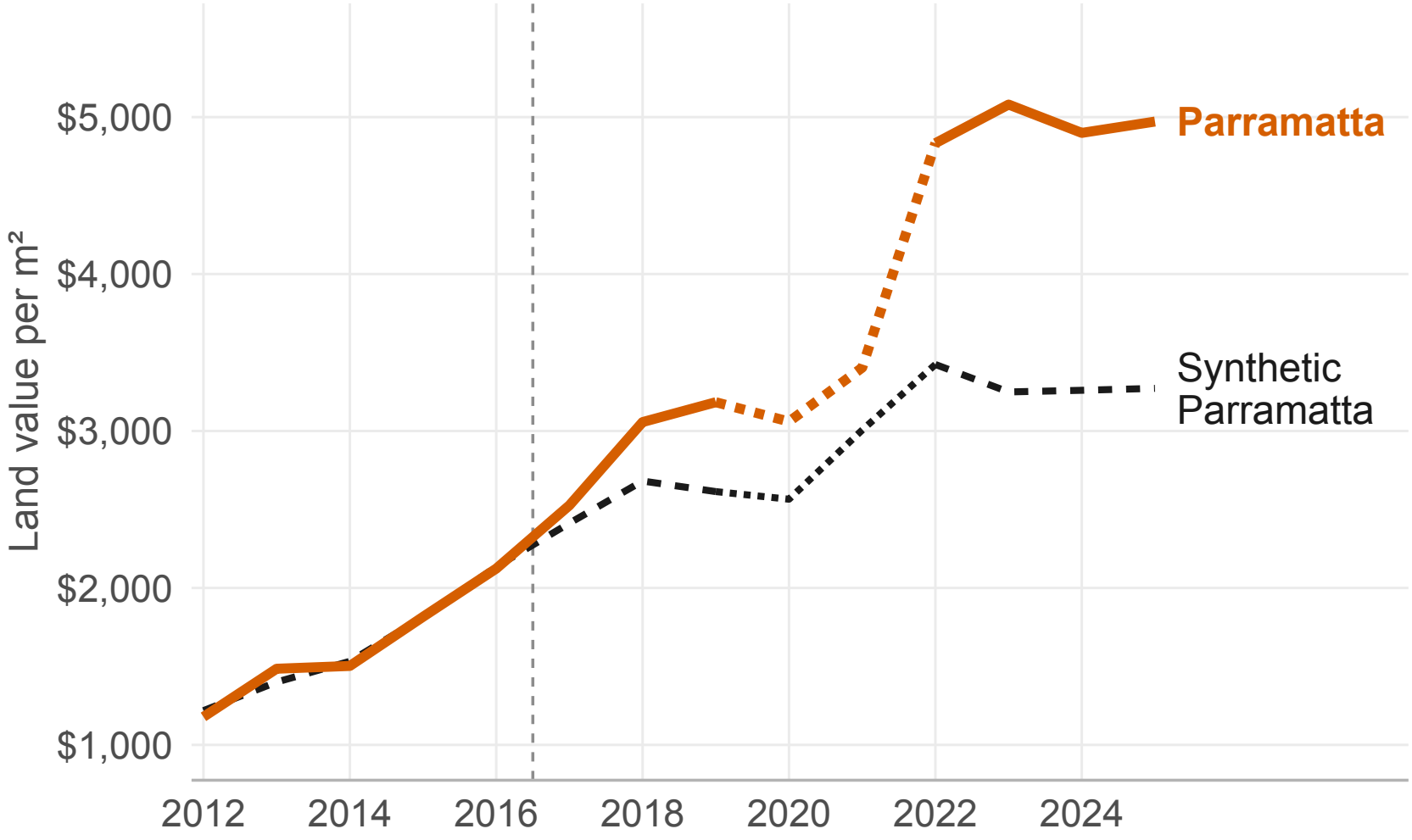
Synthetic control: four centres carry the weight



800 m circles over 2016 jobs per km^2; ■ = CBD. Weights from the 2012–16 fit.

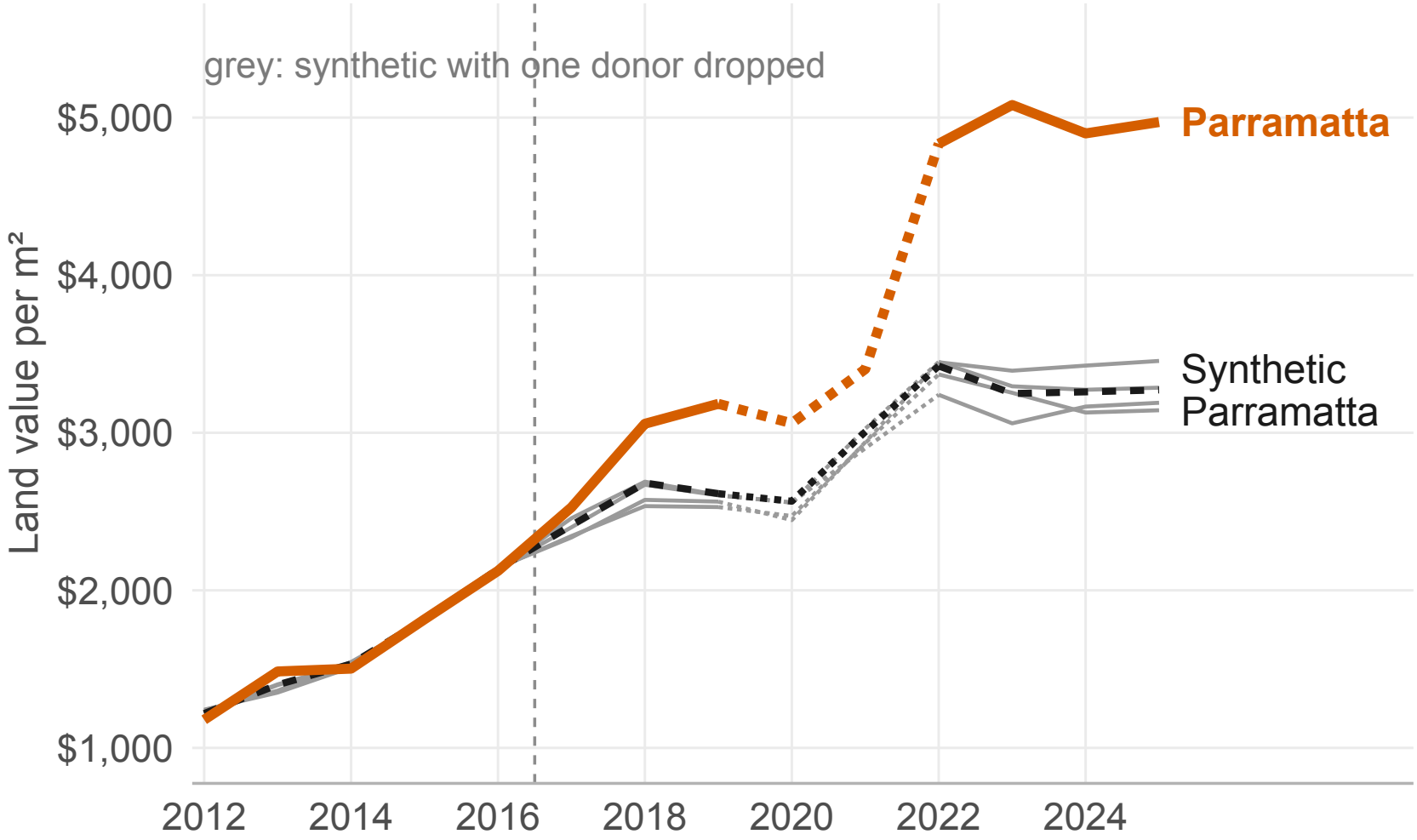
The screens →

Commercial land values pull away after the lease: +$1,700 per m^2



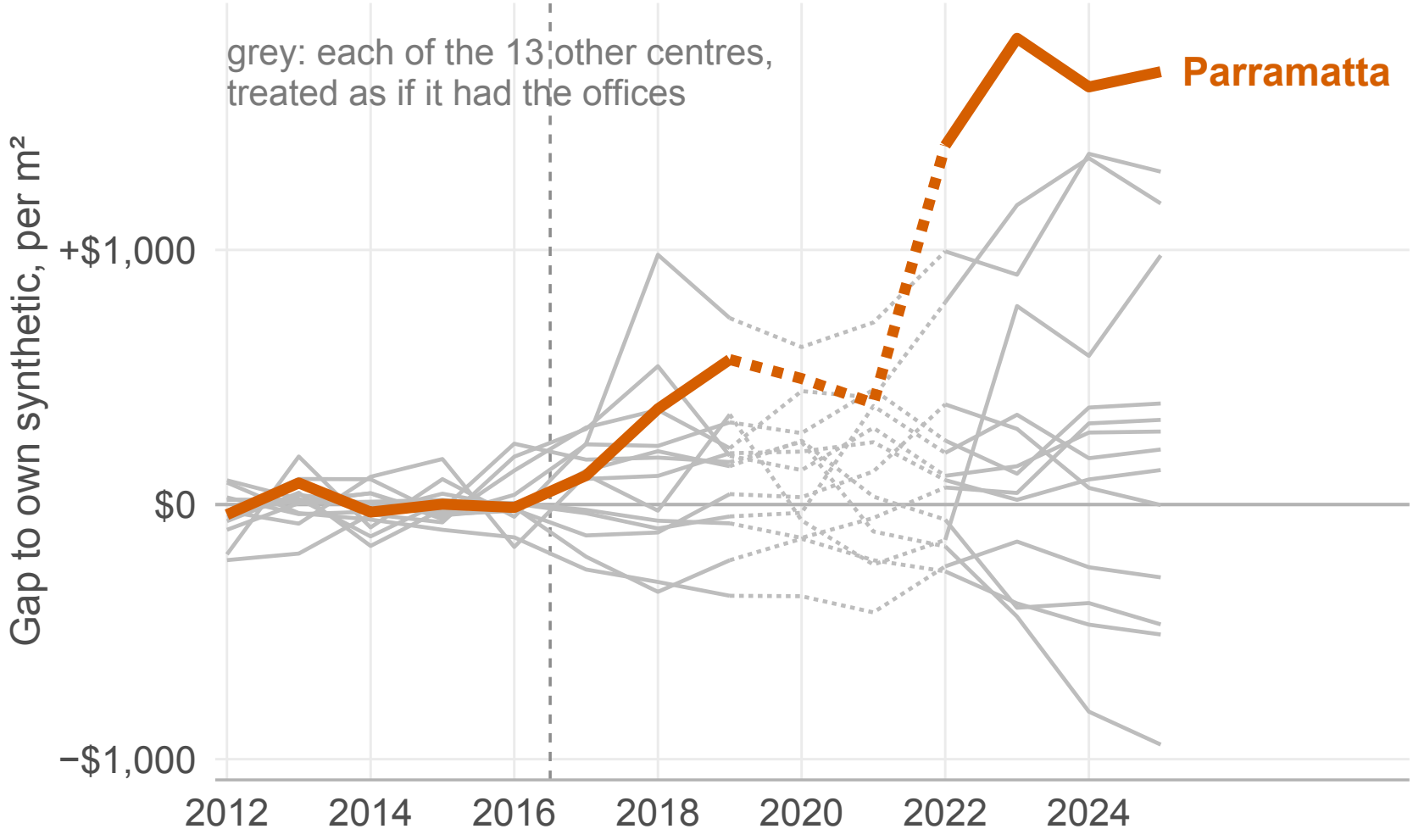
Land value per m^2 of commercial-zoned land (excl. industrial), same parcels throughout, 800 m circle. Dashed line: Dec 2016 lease.

Commercial land values pull away after the lease: +$1,700 per m^2



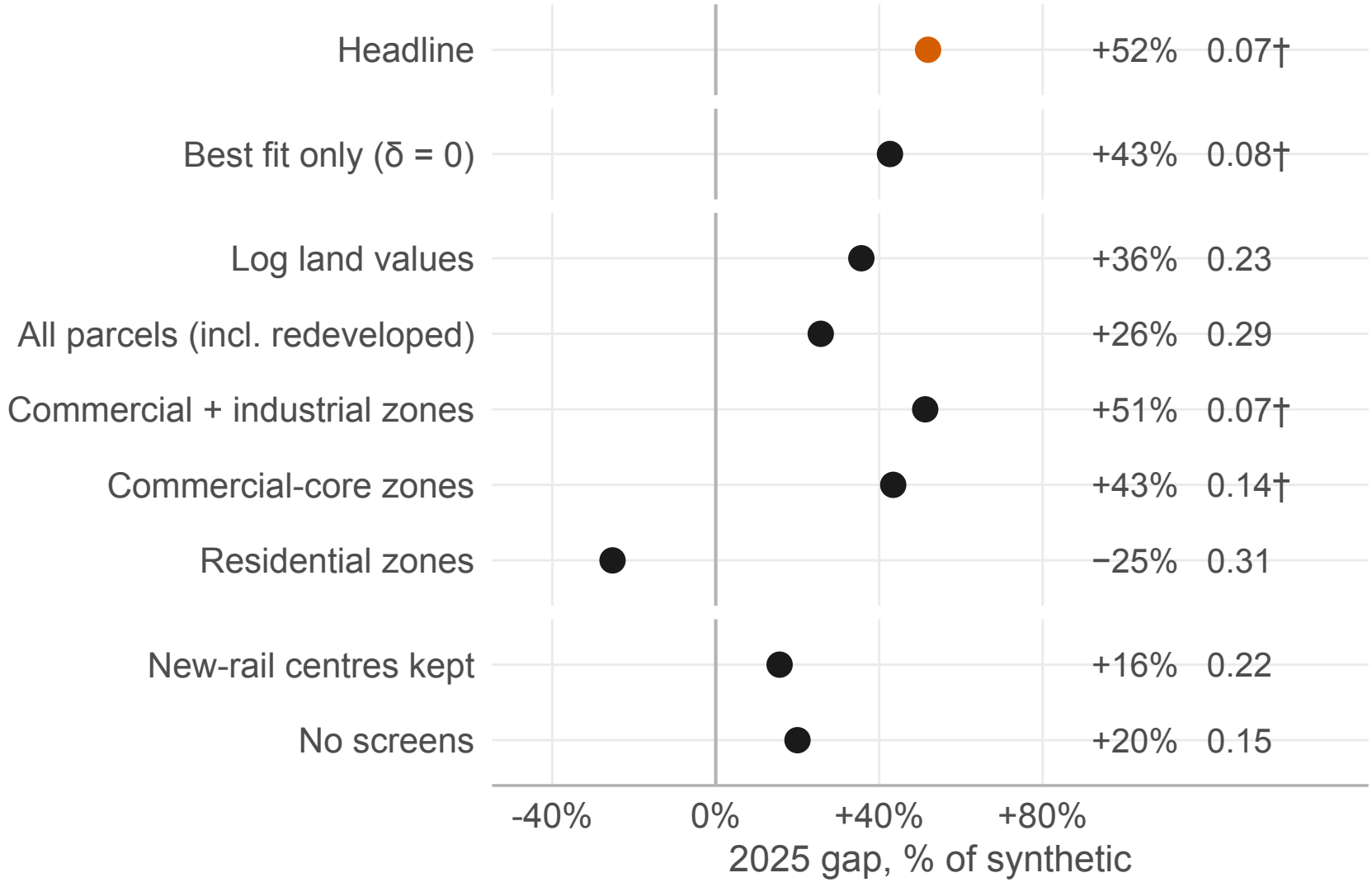
Land value per m^2 of commercial-zoned land (excl. industrial), same parcels throughout, 800 m circle. Dashed line: Dec 2016 lease. Grey: the synthetic rebuilt with each weighted donor dropped in turn.

No other centre shows a gap this large



Each grey line: one donor centre, with its own synthetic control, as if it had received the offices (the 13 that fit their own pre-period well). Parramatta's post-2017 gap is the largest: $p = 1/14 = 0.07$.

The gap is robust to the outcome — less so to the donor screens

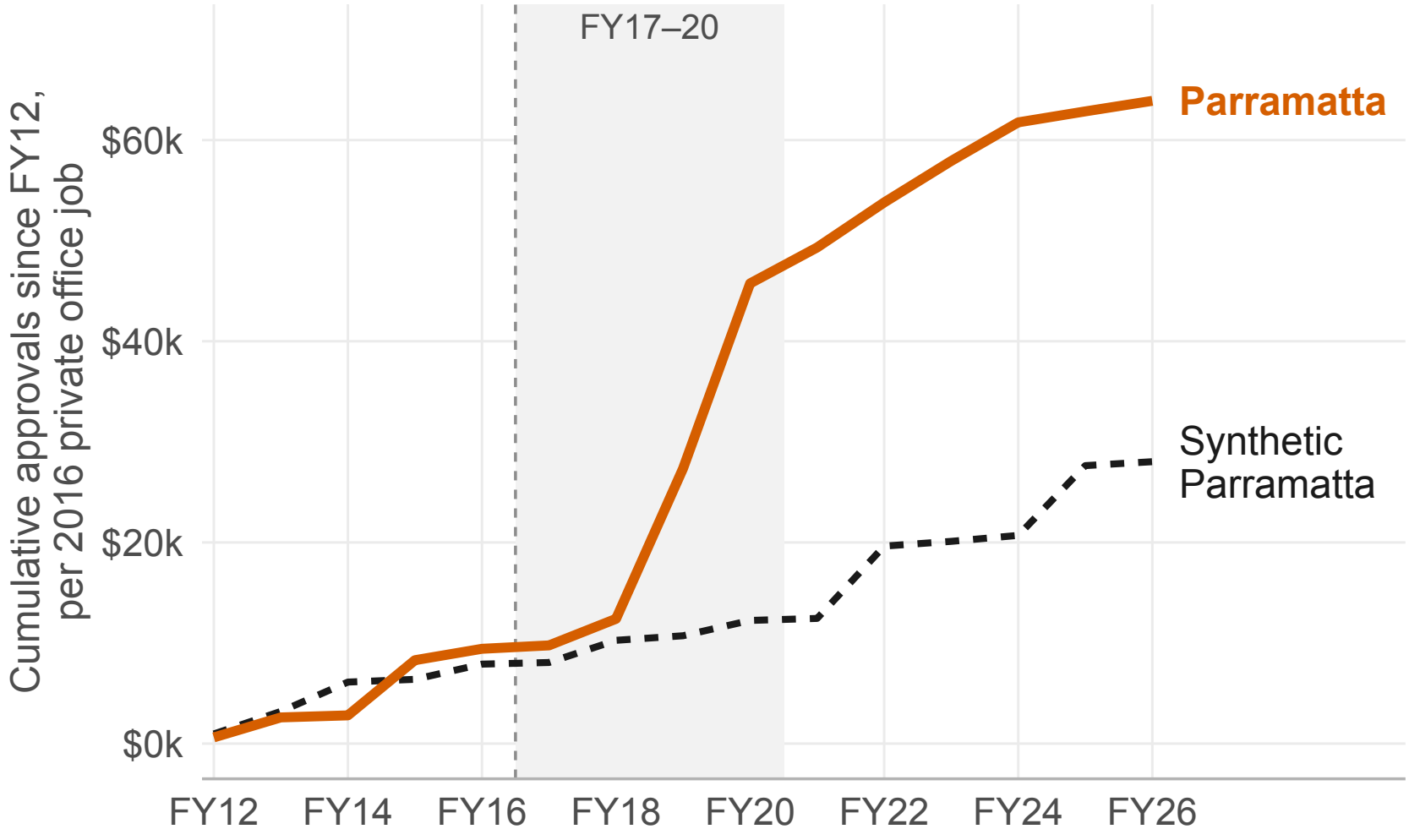


Right: 2025 gap (% of synthetic), placebo *p*; † = Parramatta's gap is the largest.

Second outcome: private office approvals

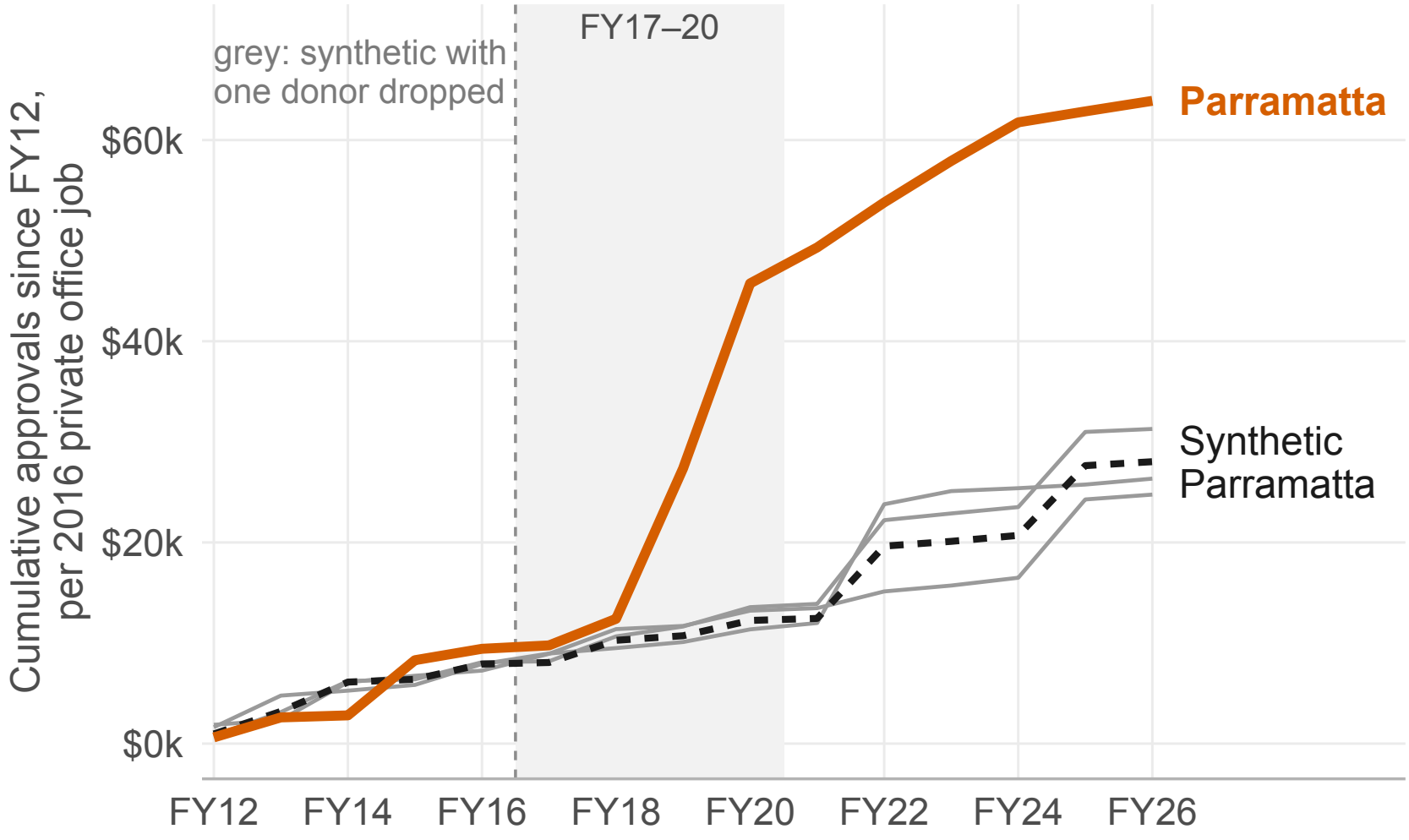
- **Data:** ABS building approvals: $ of office work by small area, monthly, 2011–26
- **Units:** regions with a stable ABS footprint; Parramatta $\approx 8.5\ \text{km}^2$
- **Private only:** ABS codes the *owner*, not the tenant
  - net out the government floors of the three leased towers
- **Outcome:** cumulative real approvals since FY2012, per 2016 private office job
- **Same estimator:** donors = the 10 regions most similar to Parramatta; fit FY2012–16; effect = gap opening over FY2017–20

Approvals pull away in FY2019–20: +$703**m**



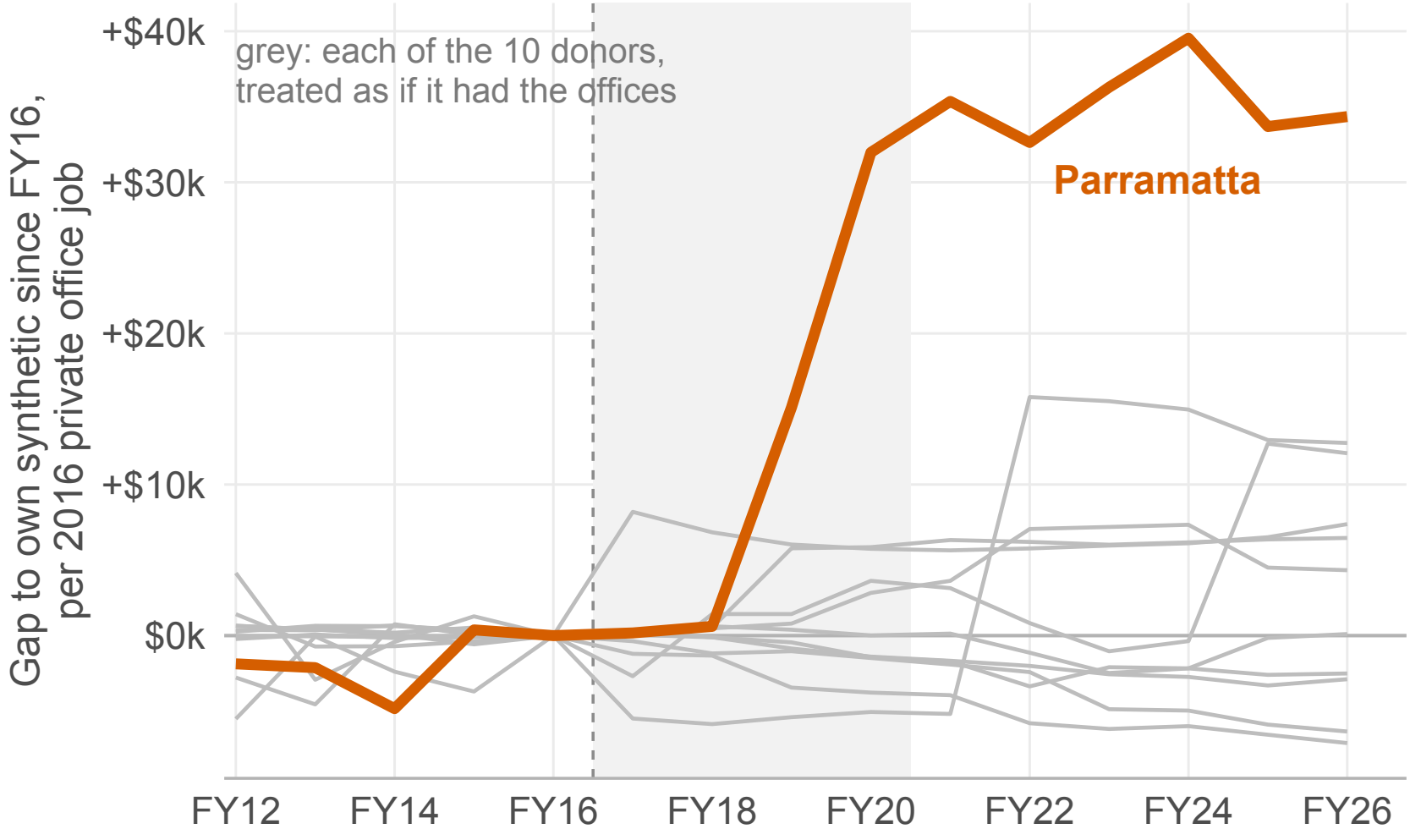
Cumulative real private office approvals (FY2016–17 prices) per 2016 private office job.
Pre-period error 38 % of Parramatta's level.

Approvals pull away in FY2019–20: +$703**m**



Cumulative real private office approvals (FY2016–17 prices) per 2016 private office job. Pre-period error 38 % of Parramatta's level. Grey: each weighted donor dropped; +$668m to +$723m.

No donor shows a gap close to Parramatta's



Gap to own synthetic, re-based at FY2016. Grey: each donor on its own pool. p = 1/11 = 0.09, the floor.

In jobs: room for about 6,900 office workers

$$\text{jobs} = \text{gap} \div \text{overshoot} \div (\$ \text{ per } m^2 \text{ GFA}) \times (\text{NLA / GFA}) \div (m^2 \text{ per worker})$$

Input	Central	Range	Source
\$ per m^2 GFA	3,732	3,101–4,296	five Parramatta towers
Approvals ÷ consent GFA	1.194	1.148–1.40	Parramatta, FY17–20
NLA / GFA	0.77	0.68–0.86	office-building band
m^2 per worker	17.7	14.5–20.7	City of Sydney floorspace survey
Jobs per \$1m	9.8	5.5–13.7	
The \$703m gap, jobs	≈ **6,900**	3,800–9,600	sector m^2: 7,500–8,400

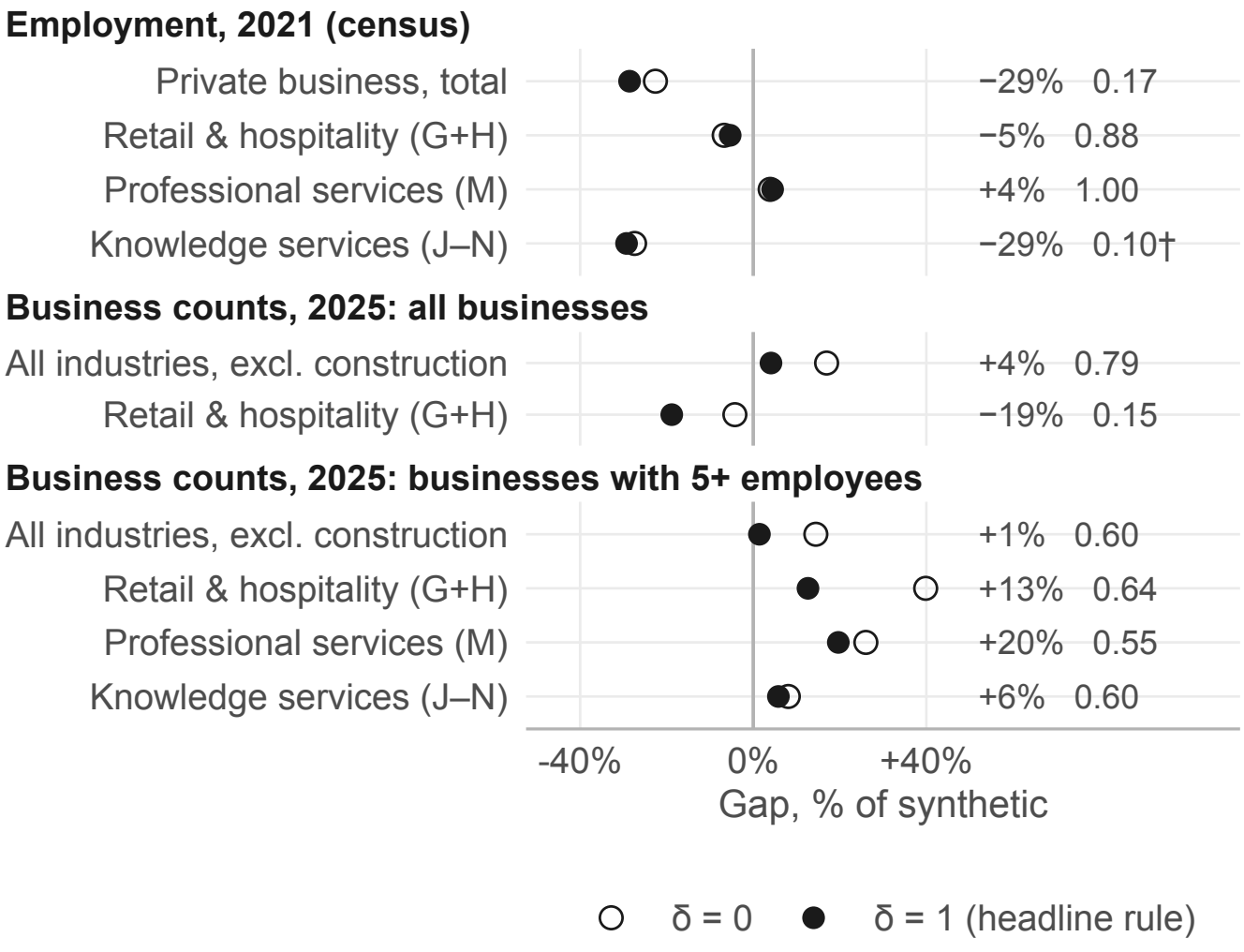
GFA: gross floor area. NLA: net lettable area, the rentable floor. Overshoot: approvals over consented GFA's value (fit-outs, repeats).

- **Capacity**, not realised jobs: consents, not floor space
- Mostly delivered after census 2021: 2026 is the test

Robustness →

Private jobs: no detectable response yet

Private jobs: no detectable response yet



Right: $\delta = 1$ gap, placebo p. Employment: 2021 Census (lockdown; before 6 PSQ). Business counts: ABS register (no government), 2025.

Dynamic version: slow re-sorting, forward-looking land prices

- Each 5-year period a share θ of workers re-choose home and job:

$$L_{ij,t+1} = (1 - \theta)\, L_{ij,t} + \theta\, L_t\, m_{ij,t+1}$$

  $L_{ij,t}$: workers living in i, working in j; m_{ij}: choice shares, given expected future utility

- Builders price floorspace off the *whole* future path of rents; floorspace depreciates

- Land values capitalise the policy on announcement (Dec 2016)

$\rightarrow$ *the static model is the dynamic model's long run*

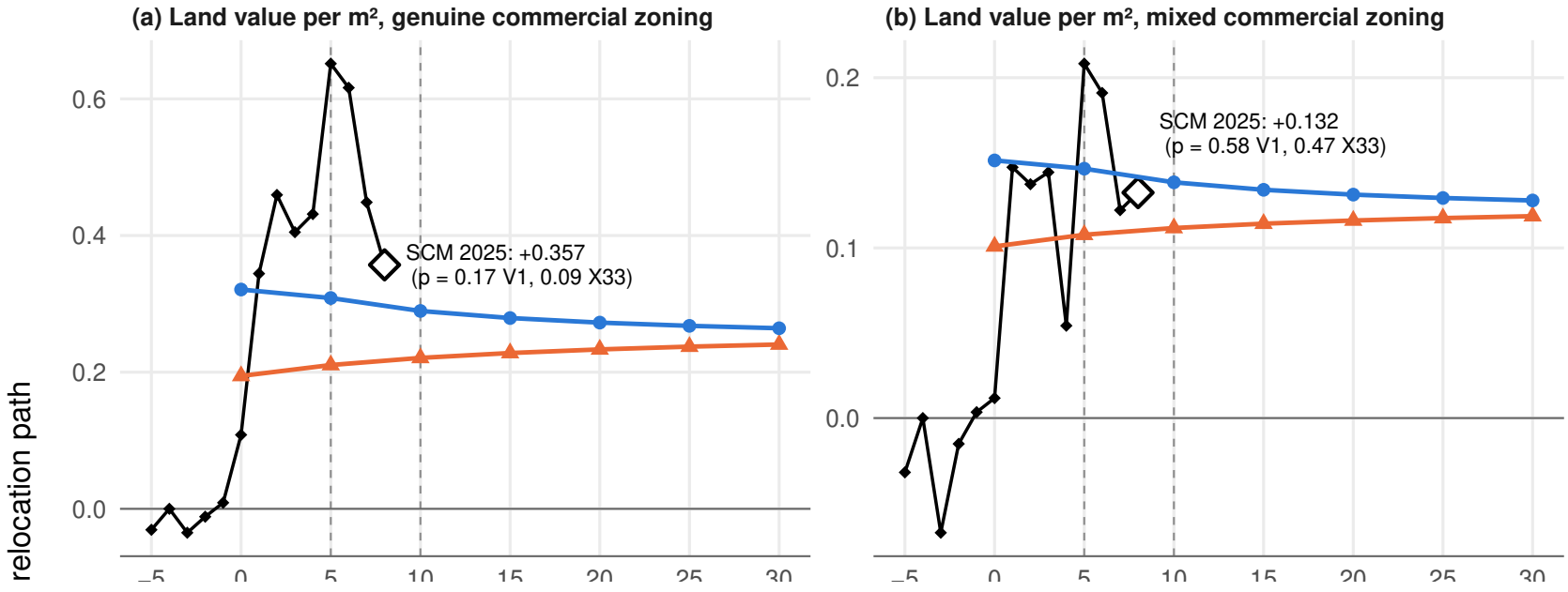
Calibration: Half of workers re-choose home and job every five years

Parameter	Value	Source
Re-choice rate θ, per 5 yrs	0.5	ACLD job moves
Depreciation, per 5 yrs	0.145	Baum-Snow & Han
Housing share, h / l	.28 / .32	expenditure surveys
Commuting decay, h / l	.112 / .134	gravity by education
Public wage, h / l	1.07 / 0.78	2016 Census

$\rightarrow$ *shock:* +9,700 *public jobs by 2021,* +4,200 *more in 2022*

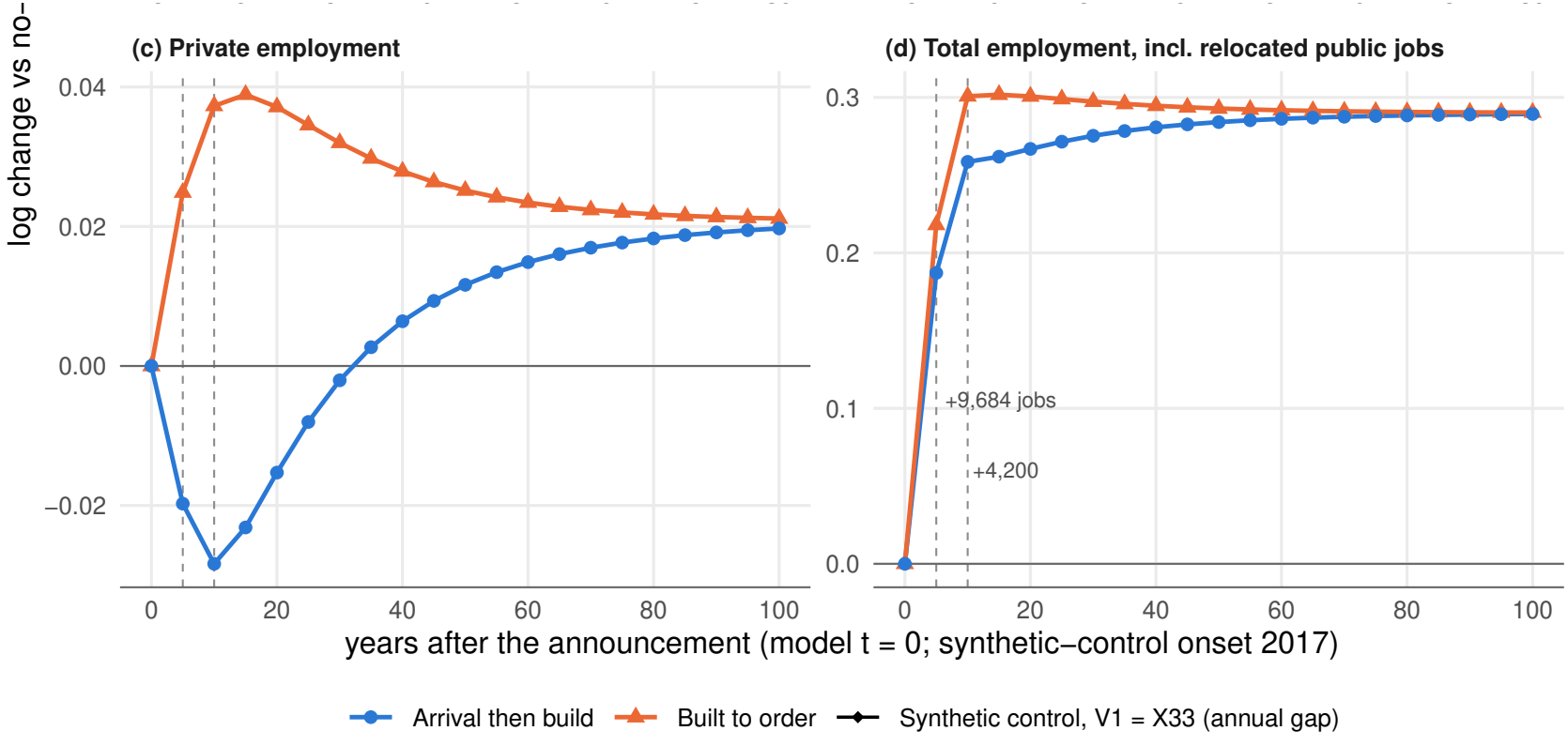
h/l: graduates / non-graduates. Public wage relative to the private wage of the same type. The dynamic runs use an earlier Fréchet dispersion (6.49) and housing shares (0.27/0.33) than the static headline.

The dynamic model: prices jump at once, private jobs follow slowly



Log change in land value vs. no relocation; years after the Dec 2016 announcement. Black: an earlier synthetic control (log, all commercial zones). Headline design above, in logs: +0.31.

The dynamic model: prices jump at once, private jobs follow slowly



Log change in private and total employment within 800 m of Parramatta Square vs. no relocation.

Conclusion

- Lopsidedness is costly: a central CBD $\Rightarrow$ output +11%, welfare +2.5%, most for graduates
- In so far as the policy is locally successful, it reduces citywide output but may help non-graduates
- Cause: concentrating employment in the CBD's graduate-intensive technology helps graduates; dispersing away from there helps non-graduates
- Data: business land values +52% vs a synthetic control; private jobs not (yet) following
- Dynamic model: prices capitalise at once, jobs follow slowly

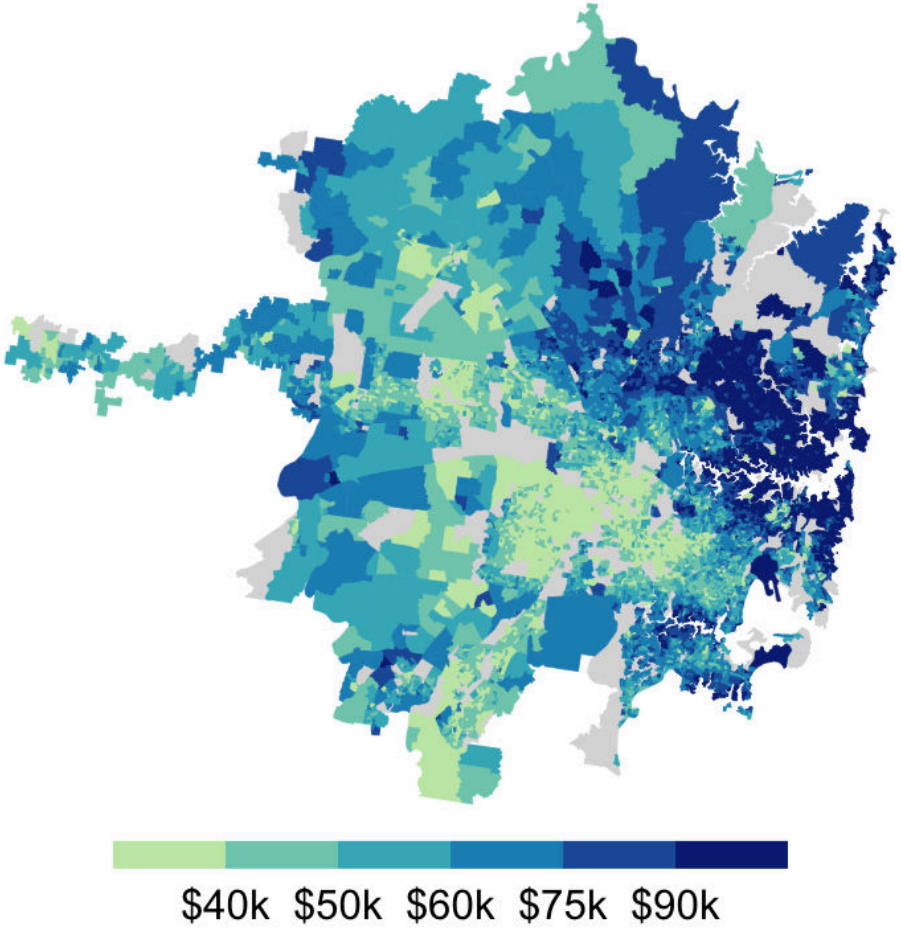
$\rightarrow$ *next: the 2026 Census, and ϕ_P from the land-value response*

Map data credits

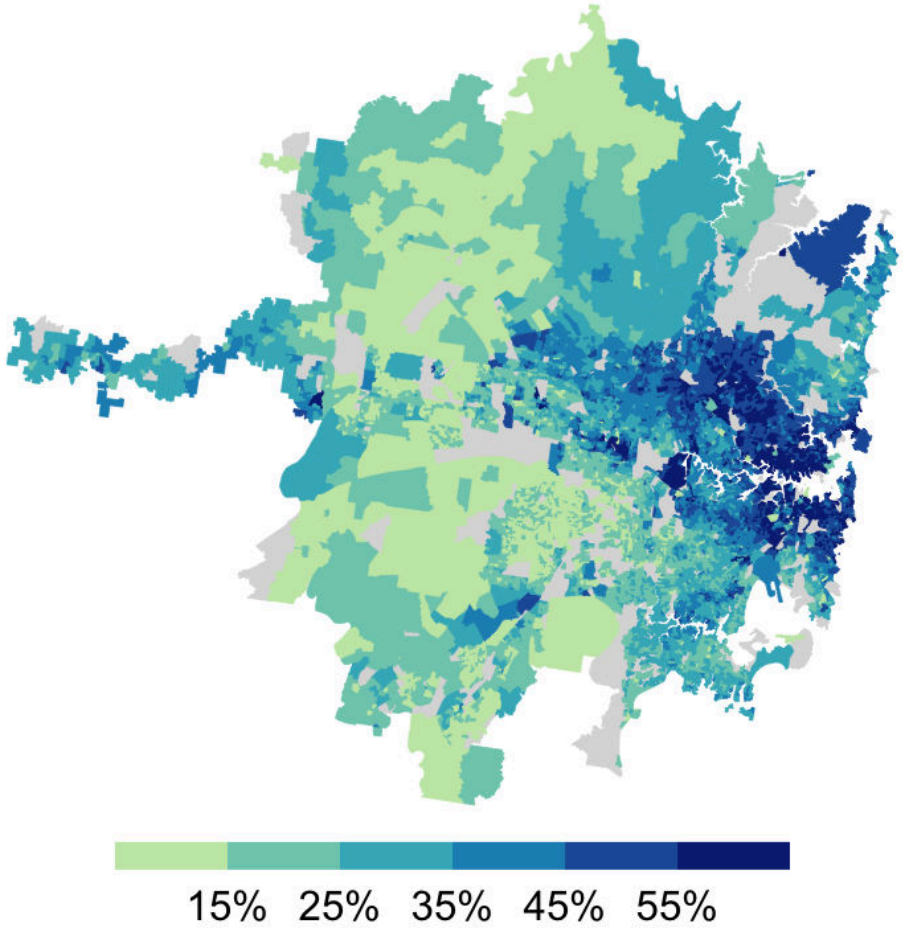
- **Satellite imagery (EOX):** EOxCloudless `https://cloudless.eox.at` by EOX IT Services GmbH (Contains modified Copernicus Sentinel data 2024), CC BY-NC-SA 4.0; colour-adjusted by the author.
- **Land cover:** © ESA WorldCover project 2021 / Contains modified Copernicus Sentinel data (2021) processed by ESA WorldCover consortium, CC BY 4.0 (Zanaga et al. 2022, doi:10.5281/zenodo.7254221); aggregated to class shares and recoloured by the author.
- **Relief:** AWS Terrain Tiles. Australia terrain data © Commonwealth of Australia (Geoscience Australia) 2017, CC BY 4.0; SRTM and GMTED2010 courtesy of the U.S. Geological Survey; ETOPO1 courtesy of NOAA.
- **Census geography and counts:** Australian Bureau of Statistics, CC BY 4.0. Land values: NSW Valuer General.

Income and graduates by home: census areas

Mean personal income



Share of graduates

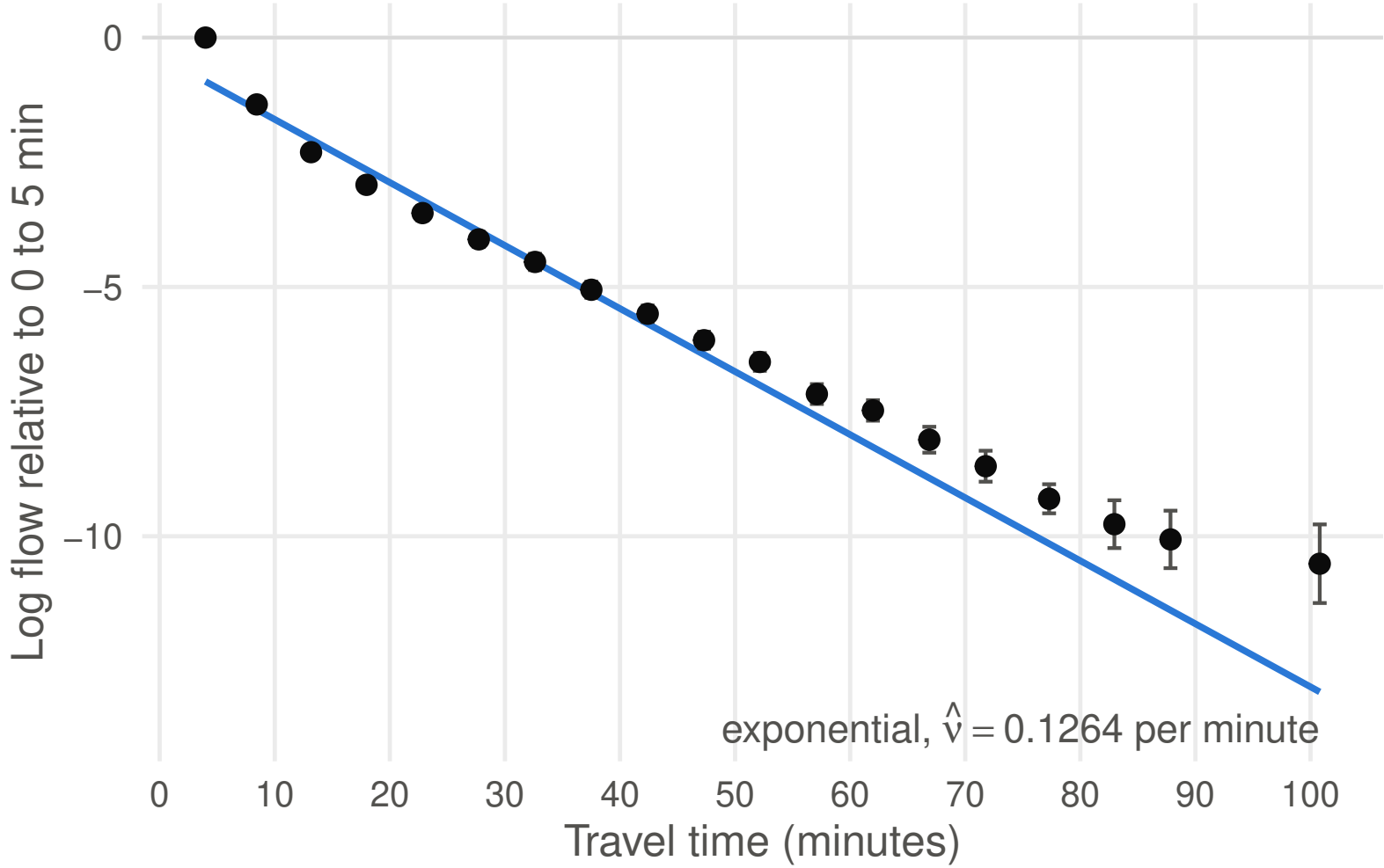


By home location, residents aged 15+. Income: residents with a positive income. Graduates: bachelor's degree or higher. Grey: under 20 residents. 2016 Census, SA1.

← Back

Backup

Commuting decays exponentially with travel time



PPML gravity on 2016 Census commuting flows (SA2 × destination zone, Sydney): $\hat{\nu}$ = 0.126 per minute (s.e. 0.002), half-life 5.5 minutes.

← Back

Backup

Calibration: Inverting the equilibrium conditions

- **Adjusted wages** $\omega_j \equiv w_j^{\varepsilon}$ clear the commuting market, given where people live and work:

$$L_{Fj} = \sum_i \frac{\omega_j \, e^{-\nu \tau_{ij}}}{\sum_s \omega_s \, e^{-\nu \tau_{is}}} L_{Ri} \qquad \Rightarrow \qquad \mathrm{Var}\left(\ln w_j\right) = \frac{1}{\varepsilon^2} \, \mathrm{Var}\left(\ln \omega_j\right) \text{ sets } \varepsilon$$

$\propto$: relative to the geometric mean. Q: floorspace price, from land values. Wage dispersion compared across SA2s: census wages are not observed by hexagon.

← Back

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- **Productivity**, from zero profits: high wages despite high rents

$$A_j \;\propto\; \omega_j^{\alpha/\varepsilon} \, Q_j^{1-\alpha}$$

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← Back Backup

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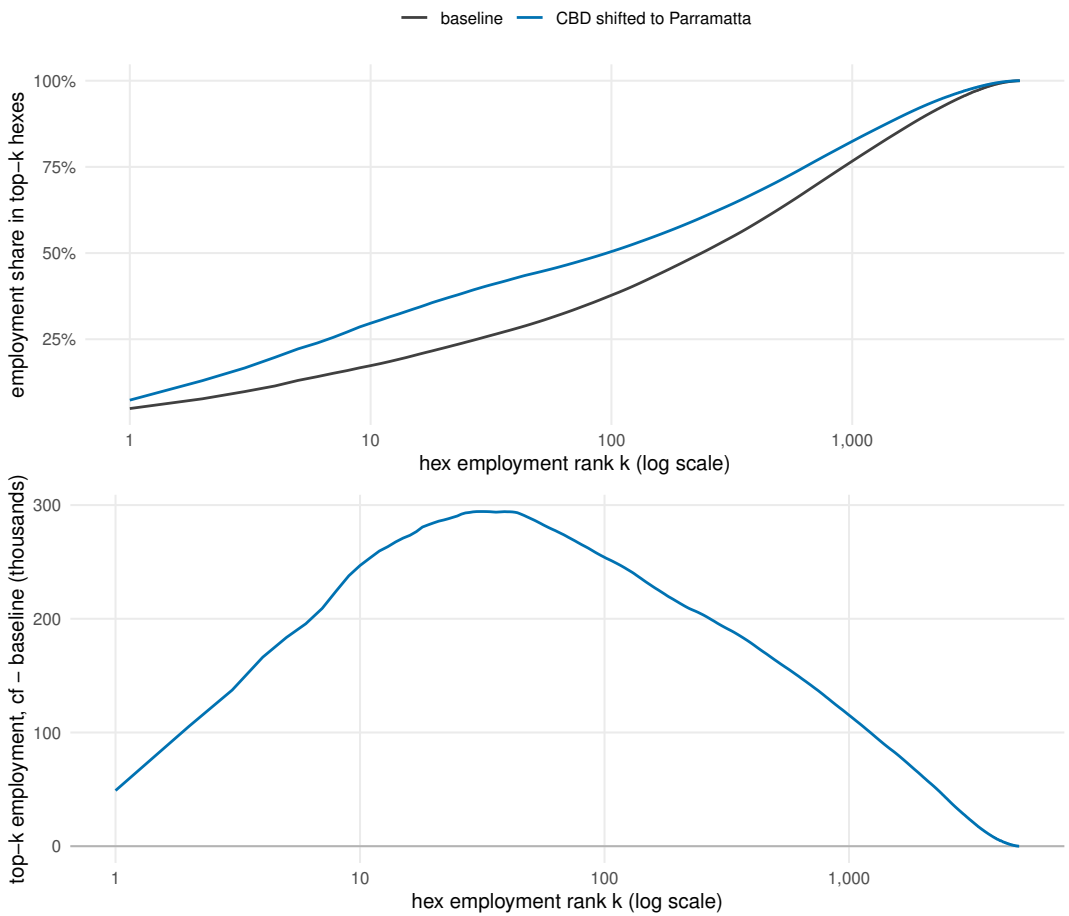
- **Amenity**, from residential choice: many residents despite high rents and poor job access

$$B_i \;\propto\; L_{Ri}^{1/\varepsilon} \, Q_i^{1-\beta} \, W_i^{-1/\varepsilon}, \qquad W_i = \sum_s \omega_s \, e^{-\nu \tau_{is}} \text{ (job access)}$$

$\propto$: relative to the geometric mean. Q: floorspace price, from land values. Wage dispersion compared across SA2s: census wages are not observed by hexagon.

← Back Backup

A central CBD concentrates employment more, not less



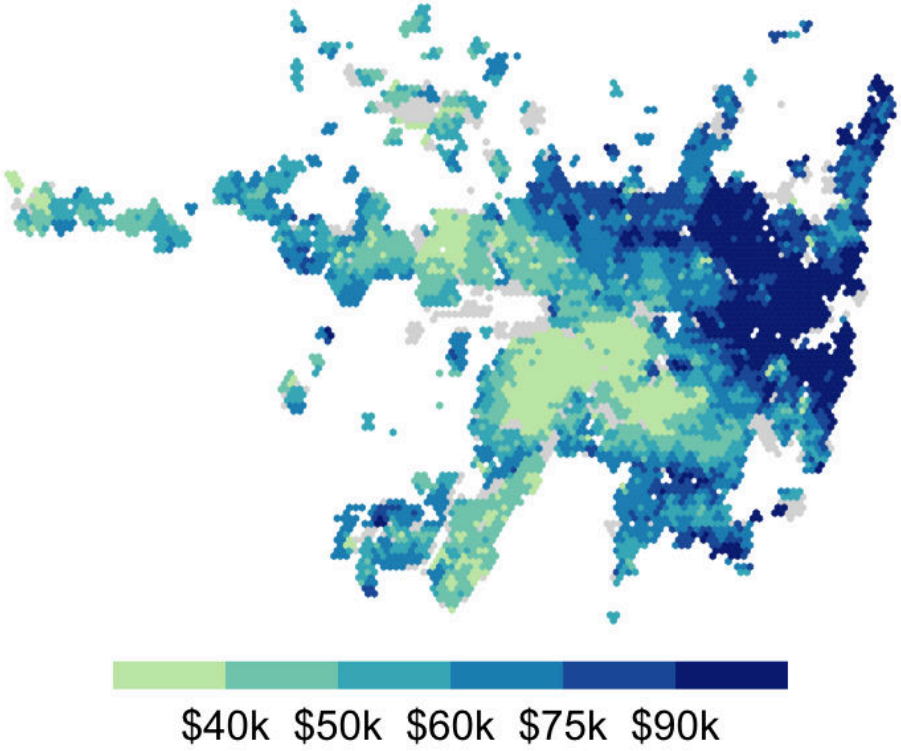
Cumulative share of jobs in each city's *k* densest hexagons. Counterfactual above baseline at every *k*; peak surplus 294,000 jobs at $k = 31$ (41% vs 26% of jobs).

← Back

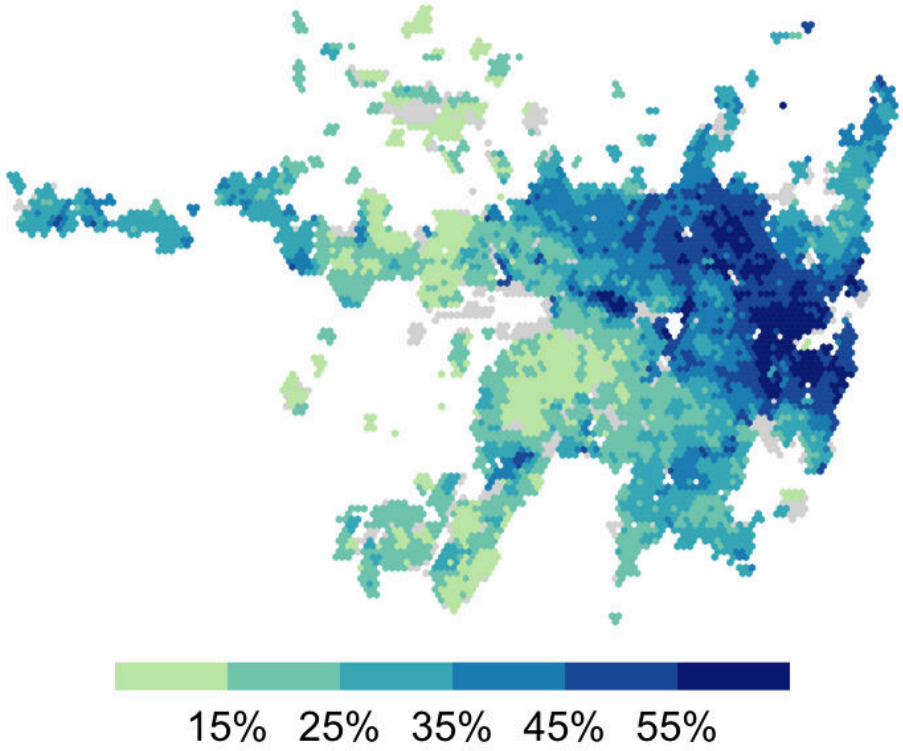
Backup

High earners and graduates live in the east

Mean personal income



Share of graduates



By home location, residents aged 15+, on the model's 400 m hexagons. Income: residents with a positive income. Graduates: bachelor's degree or higher. Grey: under 20 residents. 2016 Census.

← Back Backup

Two-type model: Firms hire graduates and non-graduates

- **Firms:** Produce with labour L_{Fj} and floorspace H_{Fj}, with productivity A_j, elasticity of substitution σ, and graduate intensity μ_j^h

$$Y_j = A_j L_{Fj}^{\alpha} H_{Fj}^{1-\alpha}, \qquad L_{Fj} = \left[(L_{Fj}^{l})^{\frac{\sigma-1}{\sigma}} + \mu_j^h \, (L_{Fj}^{h})^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

- L_{Fj}^h, L_{Fj}^l: graduate and non-graduate workers in j. σ = 1.4.

← Back Backup

Two-type model

	Δ welfare		Δ	Δ commute	
Responses allowed	grad.	non-grad.	output	grad.	non-grad.
Jobs move; nobody changes home or job	−5.07%	−2.47%	0.00%	+17.77%	+7.93%

Two-type model, σ = 1.4. Each row adds one response to the row above; the last row is the full model. Welfare: each type's expected utility (all workers). Commute: each type's mean commute (baseline 21.0 / 21.1 min).

← Back Backup

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+ people re-choose home and job	+1.27%	−2.04%	+2.34%	+0.13%	−1.74%

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+ building supply responds	+2.17%	−0.79%	+2.94%	+0.26%	−1.36%

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+ productivity spillovers	+2.22%	−0.58%	+3.93%	+0.79%	−1.14%

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+ amenity spillovers, half strength	+2.56%	−0.40%	+4.68%	+0.40%	−1.11%

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← Back Backup

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+ amenity spillovers, half strength	+2.56%	−0.40%	+4.68%	+0.40%	−1.11%
+ amenity spillovers, full strength	+3.73%	−0.09%	+6.44%	−1.79%	−1.61%

→ graduates gain at every step; non-graduates lose a little even in the full model, though their commutes shorten throughout

Two-type model, σ = 1.4. Each row adds one response to the row above; the last row is the full model. Welfare: each type's expected utility (all workers). Commute: each type's mean commute (baseline 21.0 / 21.1 min).

← Back Backup

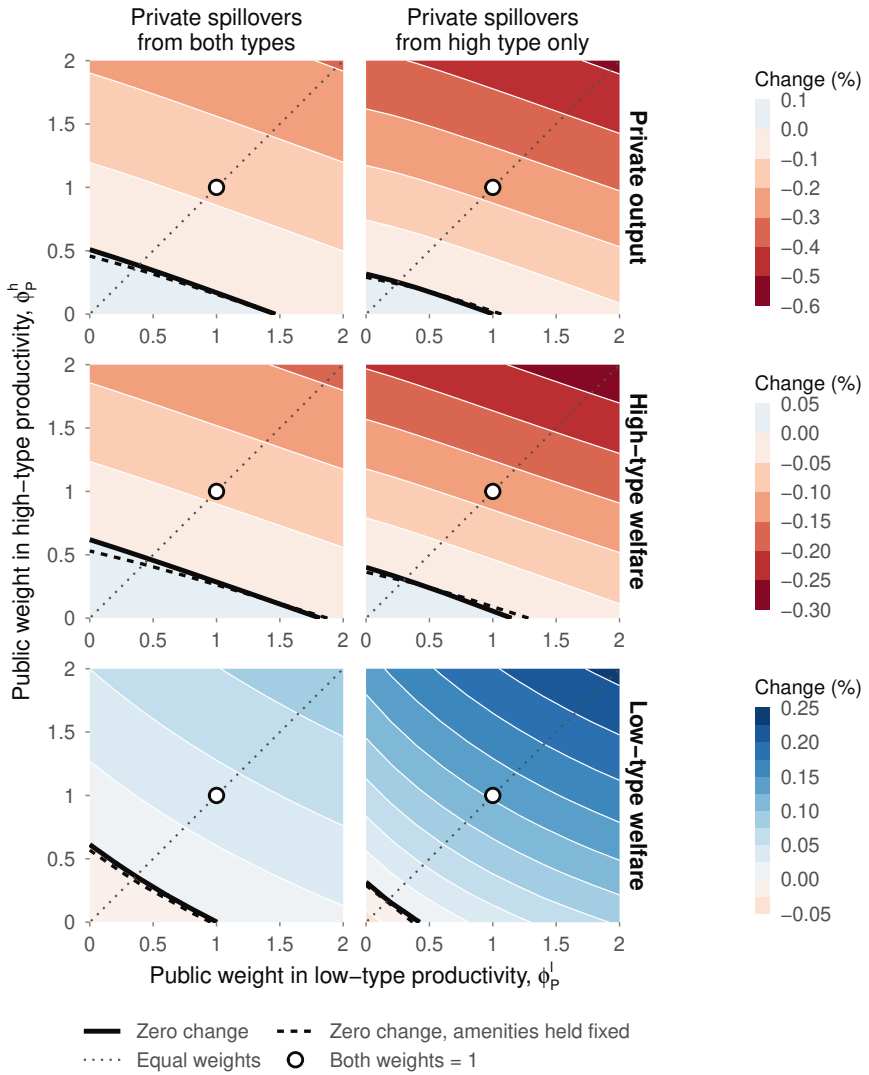
CBD shift: robustness

	Δ welfare	Δ output
Headline	+2.49%	+10.99%
Floorspace supply elasticity $\zeta = 0.5$	+0.69%	+5.93%
Floorspace supply elasticity $\zeta = 3.0$	+3.98%	+17.13%
Fréchet dispersion $\varepsilon = 4.5$	+2.64%	+13.47%
Fréchet dispersion $\varepsilon = 6.6$	+2.50%	+10.69%

Each parameter variation recalibrates all location fundamentals.

← Back

Long-run effects across the spillover weights



← Back

Backup

Which half of the spillover field drives the result

Change in non-graduate welfare (%), relative to the move with no spillover ($\phi_P^h = \phi_P^l = 0$)

	$\phi_P^h = 1,\ \phi_P^l = 0$	$\phi_P^h = 0,\ \phi_P^l = 1$
Gain at Parramatta only	+0.015	+0.029
Loss at the CBD only	+0.026	−0.002
Both (the full move)	+0.041	+0.026

Each row counts the public jobs' spillover only where they arrive, only where they leave, or both. The floorspace move is the same in every cell.

← Back Backup

The donor screens

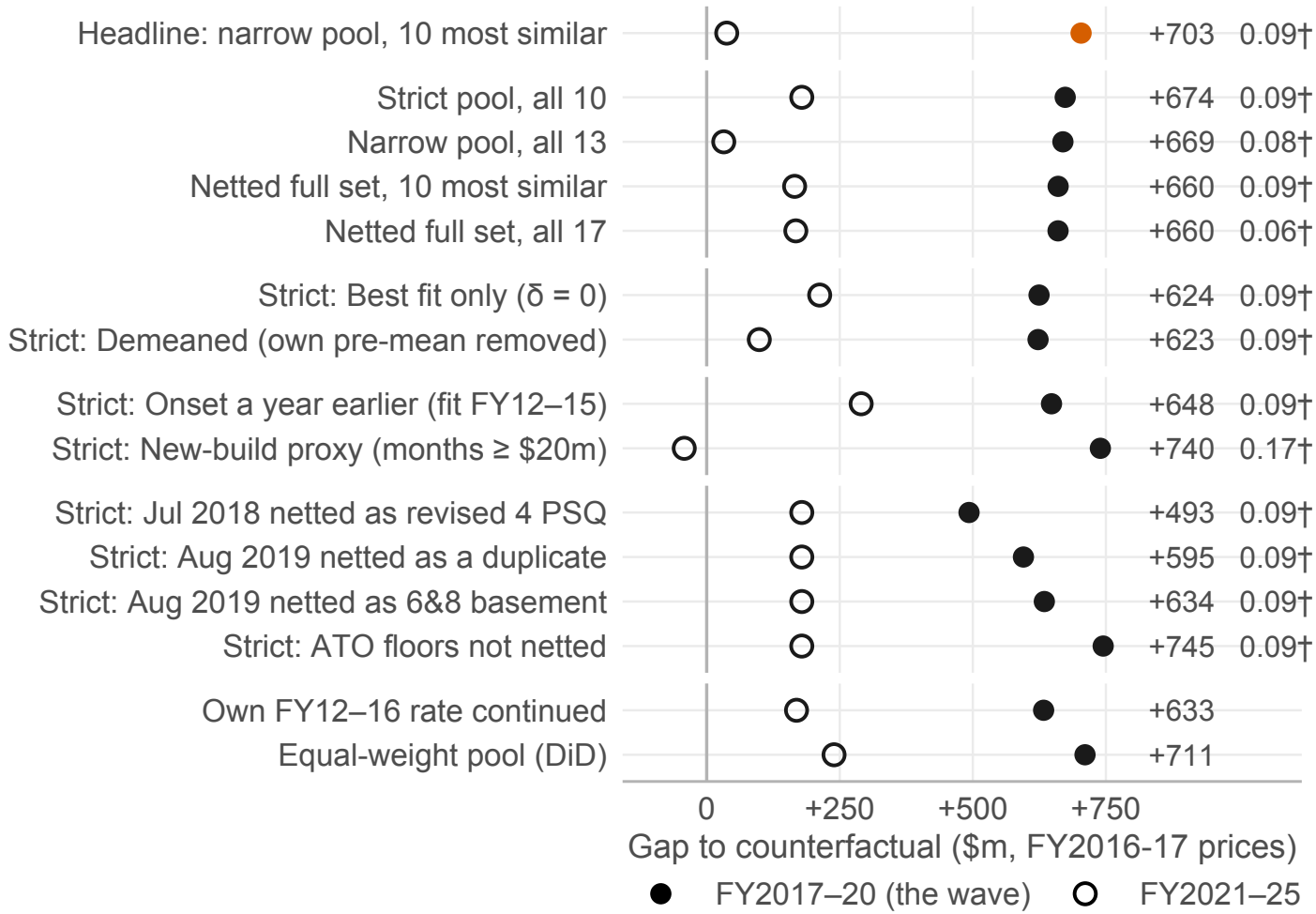
Public-jobs shock. For each centre, the 2016–21 change in government jobs as a share of its 2016 jobs, minus Sydney's (+2.25%). Screened if at least 20% of Parramatta's (+21.1%):

Centre	change in government jobs	net of Sydney
Parramatta	+9,684	+21.1%
St Leonards	+4,303	+14.1%
Campbelltown	+422	+6.1%
Liverpool	+1,287	+5.7%

New rail. A new Metro station within 1.2 km, opened 2017–25: North Sydney (Victoria Cross, 2024); Bella Vista, Norwest and Castle Hill (2019).
Tie-break ($\delta = 1$). Minimise $\sum_j w_j d_j$ (d_j: distance to Parramatta in pre-2017 characteristics) subject to the pre-period fit within 1% of the best fit; the same rule for every placebo.

← Back Backup

Approvals: robust in the wave, fragile after it



Right: FY2017–20 gap ($m), placebo *p*; † = Parramatta largest. Rule, series, netting rows: strict pool.

← Back

Backup